

The structure of the lower limb is specialized for support of the body's weight, locomotion, and maintenance of body stability (balance). Indeed, adaptations for weight-bearing and stability account for the major structural and functional differences between the upper and lower limbs. The inguinal (pelvicural) and gluteal (buttock) regions are important anatomical junctional zones between the trunk and the lower limb through which longitudinally running nerves and vessels travel (Fig. 78.1). The inguinal region includes the transitional zones between the lower limb and abdominal cavity via the myopectineal orifice (the gap between the inguinal ligament and hip joint) and inguinal canal, and provides a gateway for the passage of various structures. Similarly, the obturator canal allows for the obturator nerve and vessels to traverse between the pelvis and thigh. The gluteal region communicates with the lower limb and the abdominopelvic cavity via the greater sciatic foramen, and with the lower limb and pelvic cavity and perineum via the lesser sciatic foramen.

This chapter will provide general information on the lower limb in two sections. The first section is an overview of the general organization of the lower limb, with particular emphasis on the fasciocutaneous system, distribution of the major blood vessels and lymphatic channels, and the branches of the lumbar and sacral plexuses; it is intended to complement the detailed regional anatomy described in Chapters 80–84. The second section describes the surface anatomy of the lower

limb, which is especially important to physicians during various surgical and clinical procedures.

SKIN, FASCIA AND SOFT TISSUES

In the young adult and as an adaptation to weight-bearing, the skin of the lower limb is generally stronger and thicker than that of the upper limb. The soft tissues of the sole of the foot are particularly thickened in order to support weight during standing. The skin of the buttocks and posterior thigh bears weight in the sitting position, and consequently is relatively thick. The skin over the anteromedial aspect of the leg is particularly fragile and vulnerable in the elderly.

Subcutaneous tissue (tela subcutanea)

The subcutaneous tissue (tela subcutanea; hypodermis) of the lower limb becomes thinner peripherally. It participates in the integrity of the skin and provides support for subcutaneous structures such as superficial veins and cutaneous nerves. It is connected to the adventitia of the superficial veins by thin bands that prevent the displacement of the veins during movement. The hypodermal plexus of arteries and veins

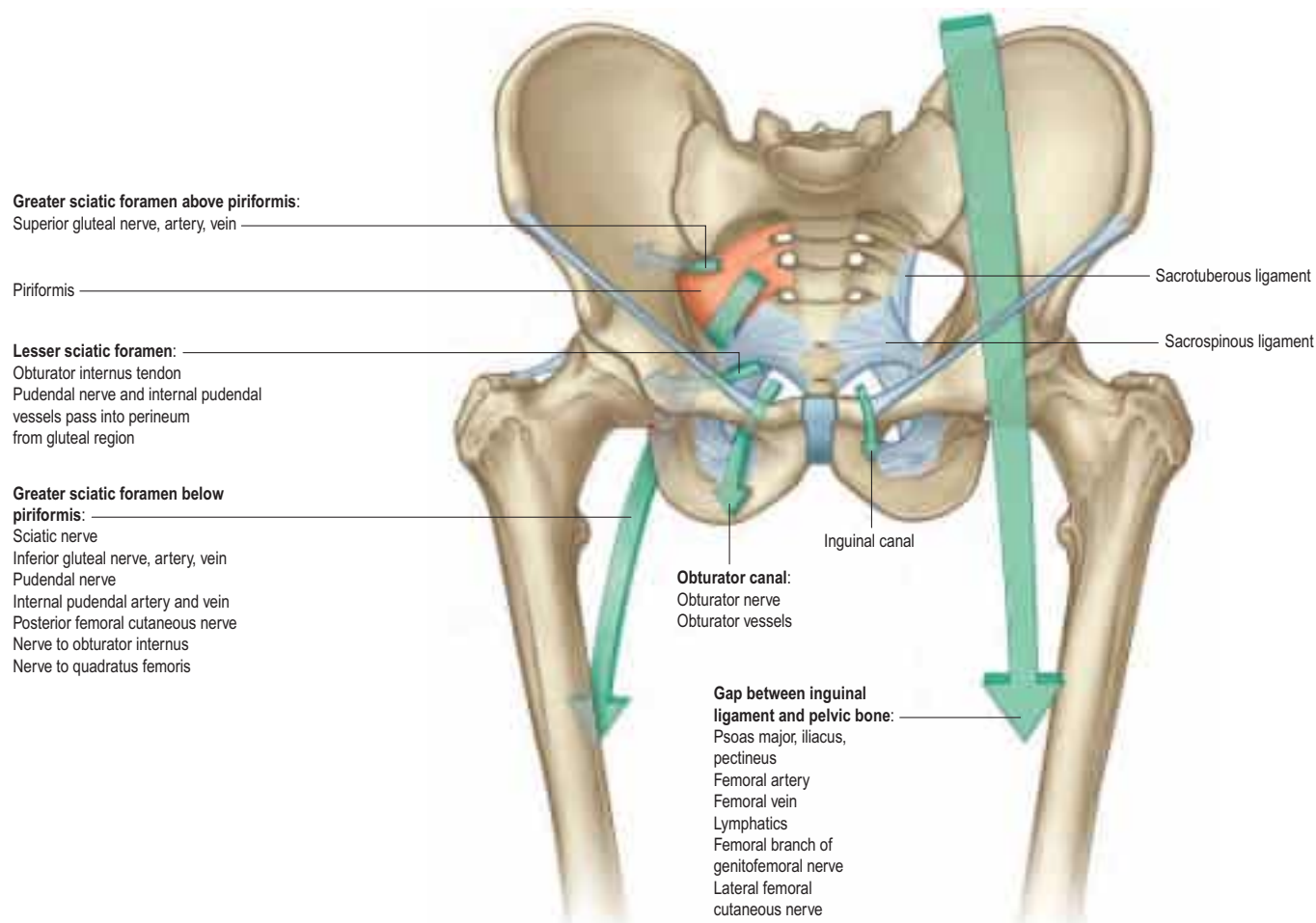


Fig. 78.1 Gateways from the abdomen, pelvis and perineum to the lower limb. (With permission from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

controls blood flow through the skin, thus helping to regulate body temperature (Sterzi 1910).

The deep fascia of the lower limb is a well-defined layer that forms a tough circumferential 'stocking-like' structure that constrains the musculature (Fig. 78.2). Septa pass from the deep surface of this fascial sheath to the bones within, confining the functional muscle groups within osteofascial compartments. The tough fascia gives additional areas of attachment to the muscles and ensures maximal function. According to Cruveilhier (1841), all aponeurotic fasciae are put under tension by specific myofascial expansions or dedicated muscles. Such thickenings may act functionally as additional tendons. Elsewhere, thickenings in the fascial skeleton form fibrous retinacula where tendons cross joints. The pattern of soft tissue organization has a bearing on the physiological effects of the muscles and is crucial for efficient venous return from the limb. The fascial planes also control and direct the spread of pathological fluids (blood, pus) within the limb and play an important part in determining the degree and direction of displacement seen in long bone fractures.

Fasciocutaneous system

The fascial septa dictate the pathways of cutaneous arteries, which subsequently perforate and ramify on the fascial 'stocking' before supplying the skin.

Osteofascial compartments in the lower limb

The muscles of the thigh may be grouped into three compartments according to their function: namely, anterior (extensor), posterior (flexor) and medial (adductor). Only the anterior and posterior compartments possess distinct fascial boundaries. A very definite fascial separation into anterior (extensor), posterior (flexor) and lateral (evertor) compartments exists in the leg, and compartment syndrome is most common in this region (see below). Osteofascial compartments in the foot are described on page 1420.

Vessels and nerves run through all the osteofascial compartments and supply the muscles contained within them. The muscles acting within these closed compartments assist in maintaining the anti-gravity flow of venous blood.

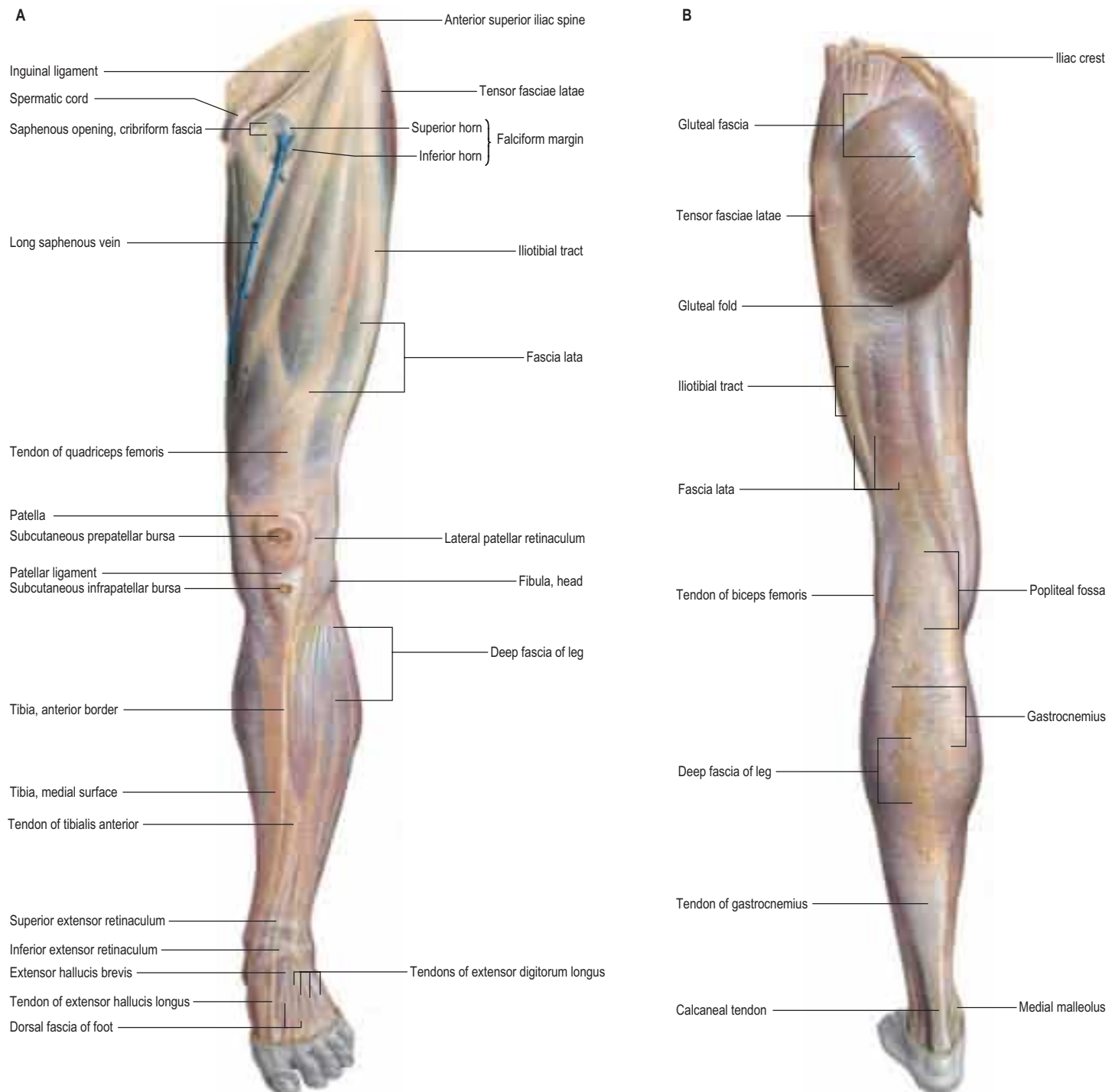


Fig. 78.2 The deep fascia of the lower limb. **A**, Anterior view. **B**, Posterior view. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

Compartment syndrome The fascial boundaries that limit the osteofascial compartments are largely inelastic, which means that any condition that leads to an increase in the volume of the compartmental contents, e.g. muscle swelling caused by trauma, haemorrhage or local infection, is likely to cause an increase in intracompartmental pressure. If unrelieved, this increased pressure will lead to compressive occlusion of the vessels in the compartment and consequent ischaemic damage to the nerves and muscles of the compartment, a phenomenon known as compartment syndrome. The normal compartmental pressure of the leg is significantly higher in children than in adults: the average pressure in the compartments varies between 13.3 and 16.6 mmHg in children, compared with 5.2 and 9.7 mmHg in adults (Staudt et al 2008).

The treatment of compartment syndrome relies on reducing abnormally elevated tissue pressure. Surgical decompression of the restricting compartment's fasciae may be necessary; in order to prevent neurovascular injury during a fasciotomy, the exact course of the nerves and vessels within the affected compartment must be known (Apyadin et al 2008).

BONES AND JOINTS

The bones of the lower limb are the three fused components of the pelvic girdle; the femur and patella (thigh); the tibia and fibula (leg); and the tarsus, metatarsus, phalanges and sesamoid bones (foot) (Fig. 78.3). The pelvic bones (especially the ilium and ischium), femur, tibia and bones of the hindfoot are strong and their external (cortical) and internal (trabecular) structure is adapted for weight-bearing.

The pelvic girdle connects the lower limb to the axial skeleton via the sacroiliac joint, a plane synovial type of joint in which mobility has been sacrificed for stability and strength, to allow for effective weight transmission from the trunk to the lower limb. Anteriorly, the pelvic girdle articulates with the contralateral girdle at the pubic symphysis, a secondary cartilaginous joint that may display a slight degree of mobility during hip and sacroiliac movement, and during childbirth. The hip joint, a synovial ball-and-socket joint, exhibits a very effective compromise between mobility and stability that allows movement in all three orthogonal planes. The more distal joints have gained mobility at the expense of stability. The knee joint is a bicompartamental synovial articulation, which occurs mainly between the femur and the tibia, and allows flexion, extension and some medial and lateral rotation of the leg. It is not a true hinge joint because its axes of flexion and extension are variable and there is coupled rotation. The knee joint also includes the articulation between the patella and femur. The tibia and fibula articulate with each other at the superior and inferior tibiofibular joints. The superior joint, a plane synovial joint, allows slight gliding movement only. The inferior joint, a fibrous joint, lies just above the ankle and allows a degree of fibular rotation linked to ankle motion. The ankle (talocrural) joint is formed by the distal ends of the tibia and fibula 'gripping' the talus, and allows dorsiflexion and plantar flexion. There are multiple joints in the foot that may be classified topographically on the basis of whether they are in the hindfoot, midfoot or forefoot. Collectively, these joints allow the complex movements required as the foot fulfils its functional roles as a platform for standing and for shock absorption and propulsion in gait.

Both the knee and ankle are commonly subject to closed injuries, and the relatively superficial location of the knee renders it susceptible to open injury and infection. Although the ankle is frequently injured and is a major load-bearing joint, the incidence of clinically significant degenerative arthritis is surprisingly low when compared with that found in the hip and knee joints.

MUSCLES

The effects of extension and medial rotation of the limb that occur during fetal development are manifest in the relative positions of the muscle groups in the thigh and the leg, and in the adult pattern of segmental cutaneous innervation (dermatomes). The role of the muscles of the lower limb in the maintenance of equilibrium during locomotion and in stance is rarely emphasized sufficiently. Many of the muscles act frequently or predominantly from their distal attachments. During both stance and locomotion, the distal attachment is often fixed and the proximal attachment is mobile, e.g. the predominant action of gluteus medius is as a pelvic stabilizer rather than as a hip abductor. In contrast, in the upper limb, the proximal muscle attachments are usually fixed and the distal attachments are mobile, an arrangement that is consistent with the prehensile function of the hand. The lower

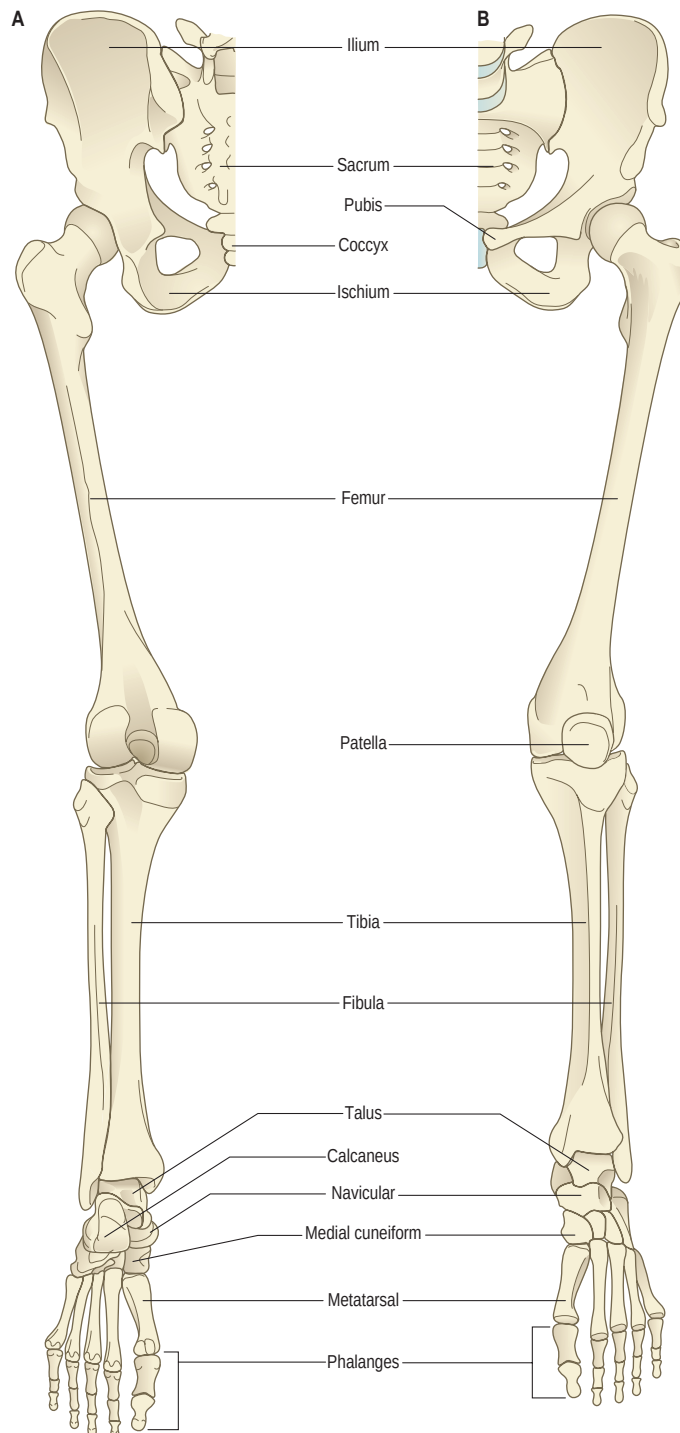


Fig. 78.3 An overview of the bones of the lower limb. **A**, Posterior aspect. **B**, Anterior aspect.

limb contains many muscles that act on more than one joint, and it is unusual for any joint of the lower limb to move in isolation.

Muscles of the lower limb may be subdivided into those of the iliac and gluteal regions, and those of the thigh, leg and foot. Note that in anatomical nomenclature, 'leg' refers to that part of the lower limb between the knee and ankle. The main muscles of the posterior abdominopelvic region are psoas major and iliacus (together called iliopsoas), the major flexors of the hip that run from the lumbar spine and inner surface of the ilium, respectively, to attach distally on to the lesser trochanter of the femur. When present, the much less important psoas minor runs from the lumbar spine to the pubis. The muscles of the gluteal region include the three named gluteal muscles and the deeper short lateral rotators of the hip joint. Gluteus maximus lies most superficially, running from the posterior pelvis to the proximal femur and fascia lata. It is a powerful extensor of the hip joint, acting more often to extend the trunk on the femur than to extend the limb on the trunk. Gluteus medius and minimus, attaching proximally to the outer surface of the ilium and distally to the greater trochanter of the femur, are abductors of the hip; their most important action is to stabilize the pelvis on the femur during locomotion, and they are helped in this

function by tensor fasciae latae, a more anteriorly placed muscle that arises from the anterolateral ilium and inserts via the iliotibial tract on to the proximal tibia. Two of the short lateral rotators of the hip, piriformis and obturator internus, arise from within the pelvis, while the others, obturator externus, the gemelli and quadratus femoris, originate externally; all of these muscles are attached distally to the proximal femur.

The muscles of the thigh lie in three functional compartments. The anterior or extensor compartment includes sartorius and the quadriceps femoris. Sartorius and rectus femoris are attached proximally to the pelvis and can thus act on the hip joint as well as on the knee, whereas the vasti are attached proximally to the femoral shaft and, acting as a unit, are powerful knee extensors. The medial or adductor compartment contains the named adductor muscles and gracilis; pectineus may also be included. These muscles are attached proximally to the anterior aspect of the pelvis, and distally to the femur; gracilis has no femoral attachment, being attached distally to the proximal tibia, while a part of adductor magnus has a proximal attachment to the ischial tuberosity. The posterior compartment includes semitendinosus, semimembranosus and biceps femoris. These muscles are attached proximally to the ischial tuberosity and act both to extend the trunk on the femur and to flex and rotate the knee. Adductor magnus, as may be inferred from the extent of its proximal attachment and its dual innervation, shares the first of these functions with the hamstrings. Biceps femoris is the only muscle of the thigh that is attached distally to the fibula, and has no tibial attachment.

In the leg, the anterior or extensor compartment includes the extensors (dorsiflexors) of the foot and the extrinsic extensors of the toes. Tibialis anterior, the main foot dorsiflexor, also inverts the foot at the subtalar joint, while the smallest muscle of the compartment, fibularis (peroneus) tertius, is a dorsiflexor that everts the foot. The posterior or flexor (plantar flexor) compartment has superficial and deep components. The superficial component contains gastrocnemius and soleus, powerful plantar flexors of the foot, and the small plantaris with its long slender tendon. Gastrocnemius and soleus are attached distally to the foot via the calcaneal (Achilles) tendon. The deep component of the flexor compartment contains popliteus, a rotator of the knee; the extrinsic flexors of the toes; and tibialis posterior, the main inverter of the foot. The lateral compartment contains the main evertors of the foot, fibularis (peroneus) longus and brevis; both muscles are also plantar flexors of the foot. Gastrocnemius and plantaris are attached proximally to the femur and distally to the calcaneus; these two muscles can therefore act on the knee as well as at the ankle. The intrinsic muscles of the sole of the foot are arranged in four layers. They facilitate the actions of the extrinsic flexors of the toes, and provide subtle changes in the shape of the foot, thereby contributing to the control of foot posture in stance and locomotion.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

ARTERIES

The femoral artery (the continuation of the external iliac artery) provides the principal arterial supply to the lower limb distal to the inguinal ligament and the gluteal fold (Figs 78.4–78.5). The femoral artery courses within the subsartorial (adductor) canal, which is located on the anteromedial aspect of the thigh. It passes through the adductor hiatus to become the popliteal artery on entering the posterior compartment of the thigh and soon thereafter divides into the anterior and posterior tibial arteries. The obturator and inferior gluteal vessels also contribute to the supply of the proximal part of the limb. In the embryo, the inferior gluteal artery supplied the main axial artery of the limb, which is represented in the adult by the artery to the sciatic nerve.

The bones of the lower limb receive their arterial supply from nutrient vessels, metaphyseal arterial branches of the peri-articular anastomoses, and the arteries supplying the muscles that attach to their periosteum. The pattern of arterial supply is particularly relevant to fracture healing, the spread of infection and malignancy, and the planning of reconstructive surgical procedures. For further details, consult Cormack and Lamberty (1984), Taylor and Razaboni (1994) and Crock (1996).

Ischaemia of the lower limb due to peripheral vascular disease is a tremendous burden on healthcare resources. Many diseases contribute to this pathology and include diabetes, hypertension and atherosclerosis. Obesity and smoking also increase the risk of developing peripheral vascular disease. Symptoms include intermittent claudication and wounds that do not heal well. Physical examination of such patients

may reveal decreased distal pulses (e.g. of the posterior tibial artery), pallor or a cool extremity.

Arterial perforators of the lower limb and surgical flaps

Achieving adequate and aesthetically satisfactory skin and soft tissue cover for large, superficial tissue defects is a perennial challenge in the field of plastic and reconstructive surgery, and accounts for a substantial part of the plastic surgeon's workload. Generally, split-thickness and full-thickness skin grafts are suitable only for very superficial defects. To achieve tissue coverage for deeper and larger tissue defects, the plastic surgeon employs one of a variety of autologous tissue flaps. A fasciocutaneous flap is composed of skin, fat and deep fascia (fascia musculorum); a lower-extremity fasciocutaneous flap (Pontén 1981) is very useful in the repair of soft tissue defects of the leg.

The viability of a flap transplanted from one part of the body to another is crucially dependent on the flap's blood supply. An appreciation of the angiosome concept, coupled with technological advances in reconstructive microsurgery, has stimulated the development and use of perforator (or perforator-based) flaps. These are flaps of skin or subcutaneous tissue supplied by one or more fascial 'perforators', i.e. arteries, which reach the suprafascial plexus either directly from a source vessel, or indirectly from some other neighbouring tissue (Harry et al 2009) (Fig. 78.6). Perforator-based flaps are typically harvested with sparing of underlying muscle tissue and minimal trauma; their use is said to reduce postoperative pain, donor site morbidity and functional loss.

The lower limb is the largest donor site in the body for perforator-based flaps. Commonly used flaps include anterolateral thigh flaps, which provide a large amount of skin; superior and inferior gluteal artery perforator flaps used in breast reconstruction; vascularized fibular flaps for reconstruction of deficient bone; vascularized tensor fasciae latae flaps for tendon reconstruction; vascularized sural nerve flaps for nerve reconstruction; and the gracilis muscle flap, used in, for example, reanimation of paralysed muscle.

In the context of perforator flap surgery, the lower limb may be considered in terms of four anatomic regions: gluteal; anterior hip and thigh; knee and leg; and ankle and foot. Each lower limb accounts for approximately 23% of the total body surface area (thigh 10.5%, leg 6.5%, buttock 2.5% and foot 3.5%) and contains an average of 90 arterial perforators (Fig. 78.7).

VEINS

The veins of the lower limb can be subdivided, like those of the upper limb, into superficial and deep groups (Figs 78.8–78.9). The superficial veins lie in the subcutaneous tissue; the deep veins (deep to the deep fascia) accompany the major arteries. Valves are present in both groups, but are more numerous in the deep veins and are more numerous in the veins of the lower versus the upper limb. Venous plexuses occur within and between some of the lower limb muscles. The two principal superficial veins are the long and short saphenous veins; their numerous tributaries are mostly unnamed; however, changes in clinical practice are driving a revision of the terminology. For further reading, see Caggiati et al (2002, 2005). The long saphenous vein can be harvested and used as a graft for peripheral vascular surgery, including coronary artery bypass (Kosinski 1926).

Deep veins of the lower limbs accompany the arteries and their branches. Plantar digital veins arise from plexuses in the plantar regions of the toes, connect with dorsal digital veins and unite to form four plantar metatarsal veins. These run in the intermetatarsal spaces and connect with dorsal veins by means of perforating veins. They then connect with each other to constitute a deep plantar venous arch adjacent to the deep plantar arch. From this venous arch, medial and lateral plantar veins run near the corresponding arteries; they communicate with the long and short saphenous veins before forming the posterior tibial veins posterior to the medial malleolus. The posterior tibial veins accompany the posterior tibial artery. They receive veins from the calf muscles, especially the venous plexus in soleus, and connect with superficial veins and with the fibular veins. The latter, running with their artery, receive tributaries from soleus and from superficial veins.

The anterior tibial veins are continuations of the venae comitantes of the dorsalis pedis artery. They leave the extensor region between the tibia and fibula, pass through the proximal end of the interosseous membrane, and unite with the posterior tibial veins, at the distal border of popliteus, to form the popliteal vein.

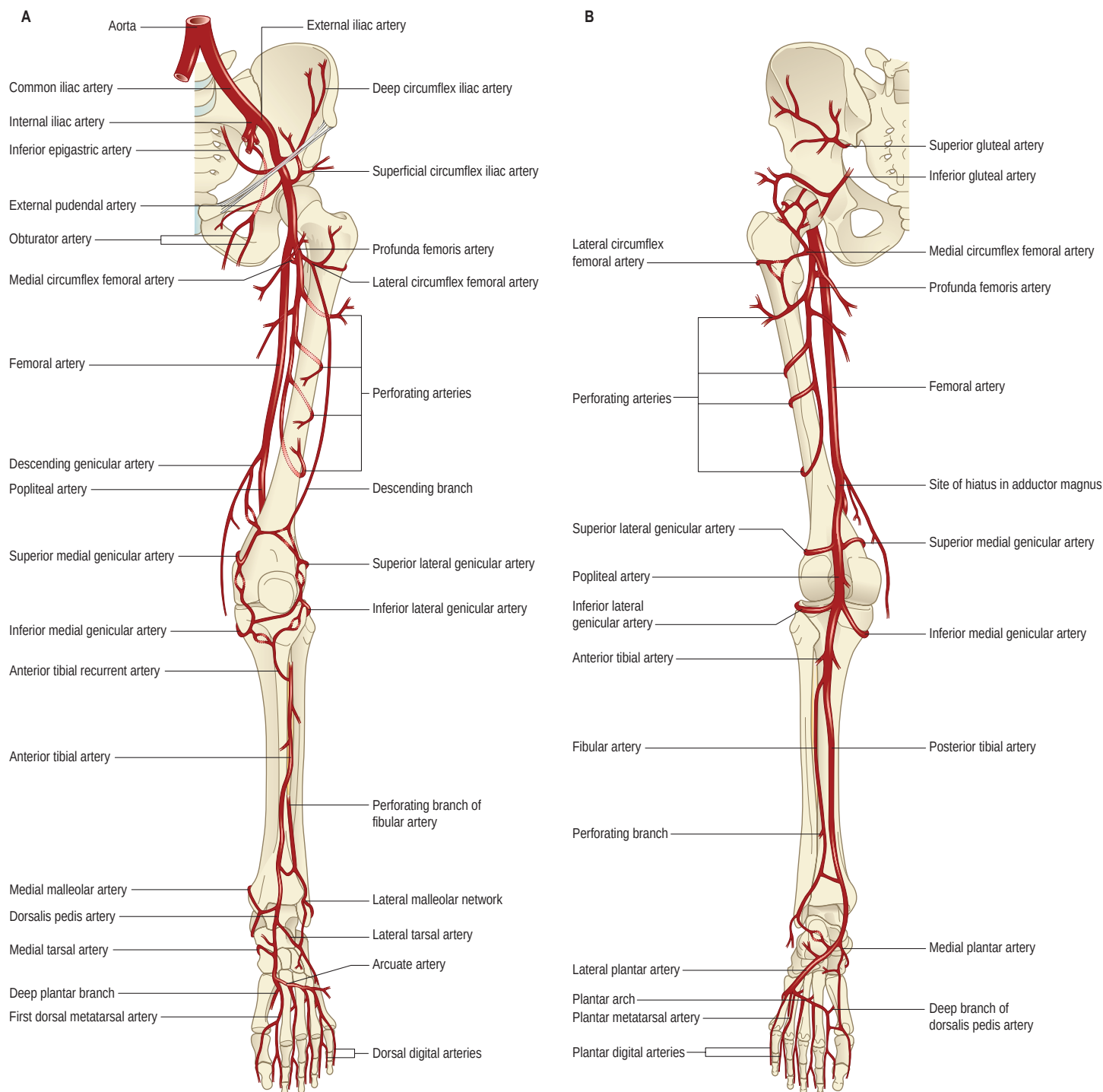


Fig. 78.4 An overview of the arteries of the lower limb. **A**, Anterior aspect. **B**, Posterior aspect.

Venous (muscle) pumps

While standing, venous return from the lower limb depends largely on muscular activity, especially contraction of the calf and foot muscles, known as the 'muscle pump', whose efficiency is aided by the tight sleeve of deep fascia (Ricci 2011). 'Perforating' veins connect the long saphenous vein with the deep veins, particularly near the ankle, distal calf and knee. Their valves are arranged so as to prevent flow of blood from the deep to the superficial veins. At rest, pressure in a superficial vein is equal to the height of the column of blood extending from that vein to the heart. When the posterior leg muscles contract, blood is pumped proximally into the deep veins and is normally prevented from flowing into the superficial veins by the valves in the perforating veins. During muscular relaxation, blood is drawn into the deep veins from the superficial veins. If the valves in the perforating veins become incompetent, these veins become sites of 'high-pressure leaks' during muscular contraction, and the superficial veins become dilated and varicose. Similar perforating connections occur in the anterolateral region, where varicosities may also occur. Incompetence of the valves

in the connecting veins between the long saphenous vein and femoral vein in the adductor canal may predispose to superficial varicosities along the medial aspect of the thigh (Dodd and Cockett 1976).

Venous plexuses

Venous plexuses may be intramuscular (soleus) or intermuscular (in the foot and gluteal region). The plexuses communicate with the axially running deep veins and are components of the 'muscle pump' mechanism.

LYMPHATIC DRAINAGE

Most lymph from the lower limb traverses a large intermediary inguinal group of nodes (Fig. 78.10). Peripheral nodes are few and all are deeply sited. Except for an inconsistent node lying proximally on the interosseous membrane near the anterior tibial vessels, they occur only in the popliteal fossa. Enlarged popliteal nodes may be palpated along the line of the popliteal vessels while the passively supported knee is

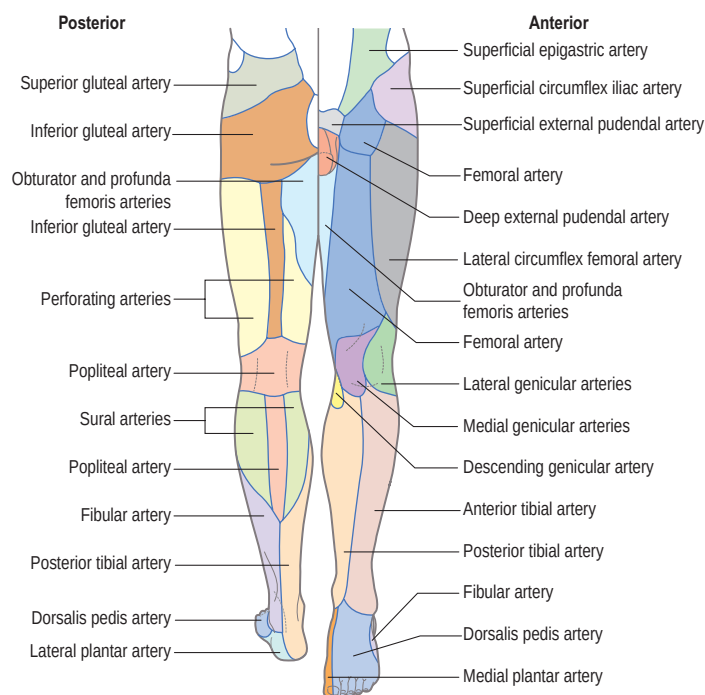


Fig. 78.5 The anatomical territories served by the cutaneous blood supply to the lower limb.

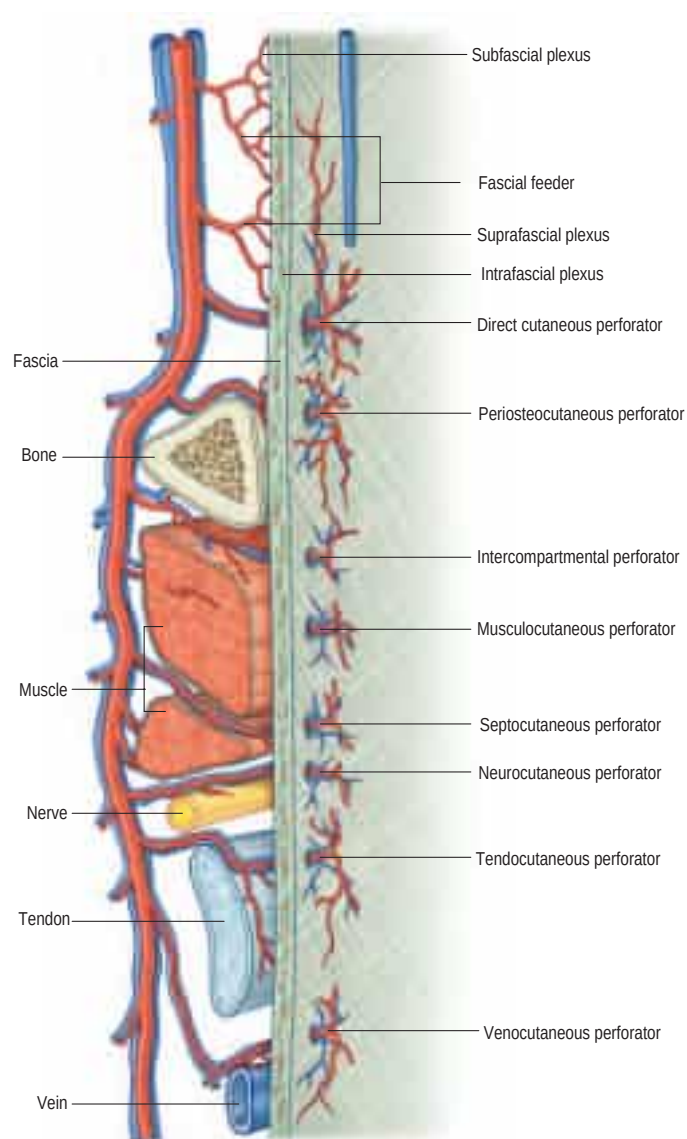


Fig. 78.6 Types of arterial perforators.

gradually moved from extension to semi-flexion. These are often due to inflammation, malignancy or injury to the lateral side of the foot. Inguinal lymph nodes are found superficial and deep to the deep fascia. The deep nodes are few and lie alongside the medial aspect of the femoral vein. The superficial nodes may be divided into a lower vertical group that clothe the proximal part of the long saphenous vein, and an upper group that lie parallel to, but below, the inguinal ligament and which are related laterally to the superficial circumflex iliac and medially to the superficial external pudendal vessels. Lymph from the lower limb passes from the inguinal lymph nodes to the external and common iliac nodes, and ultimately drains to the lateral aortic nodes. Deep gluteal lymph reaches the same group through the internal and common iliac lymphatic vessels.

Superficial tissues

The superficial lymph vessels begin in subcutaneous plexuses. Collecting vessels leave the foot medially, along the long saphenous vein, or laterally with the short saphenous vein. Medial vessels are larger and more numerous; they start on the medial side of the dorsum of the foot, ascend anterior or posterior to the medial malleolus and accompany the long saphenous vein. They drain to the distal superficial inguinal nodes. Lateral vessels begin on the fibular side of the dorsum of the foot; some cross anteromedially in the leg to join the medial vessels and so pass to the distal superficial inguinal lymph nodes, while others accompany the short saphenous vein and drain to the popliteal nodes. Superficial lymph vessels from the gluteal region run anteriorly to the proximal superficial inguinal nodes.

Deeper tissues

The deep lymph vessels accompany the anterior and posterior tibial, fibular, popliteal and femoral vessels. Popliteal nodes interrupt the deep vessels from the foot and leg; those from the thigh pass to the deep inguinal nodes. The deep lymphatic vessels of the gluteal region follow their corresponding blood vessels. Those accompanying the superior gluteal vessels end in a node near the intrapelvic part of the superior gluteal artery, adjacent to the superior border of the greater sciatic foramen, while those that follow the inferior gluteal vessels traverse one or two of the small nodes inferior to piriformis and then pass to the internal iliac nodes.

INNERVATION

OVERVIEW OF THE LUMBAR AND SACRAL PLEXUSES

Nerves derived from the lumbar and sacral plexuses innervate the lower limb (**Fig. 78.11**). The lumbar plexus lies deep within psoas major, anterior to the transverse processes of the first three lumbar vertebrae. The sacral plexus lies in the pelvis on the anterior surface of piriformis, external to the pelvic fascia, which separates it from the inferior gluteal and internal pudendal vessels. The lumbosacral trunk (L4 and L5) emerges medial to psoas major on the posterior abdominal wall and lies on the ala of the sacrum before crossing the pelvic brim to join the ventral ramus of S1. Contributions to the lumbosacral trunk may also be derived from the third lumbar nerve (**Bergman et al 1988**).

Lesions of the lumbar and sacral plexuses

The deep and protected situation of the plexuses means that lesions are not common. The lumbar plexus may be involved in retroperitoneal pathology, and the sacral plexus may be invaded by pelvic malignancies. Both may be involved in the reticulos, affected by plexiform neuroomas, or damaged in fractures of the lumbar spine and pelvis or in other conditions that cause severe retroperitoneal and pelvic haemorrhage. Temporary lesions may occur after pregnancy and childbirth, e.g. after difficult forceps delivery of a large baby. Pain, which may be diffuse, is the most common feature.

Sciatica is pain in the lower back and hip region, which radiates inferiorly along the posterior thigh to the leg. It is often caused by a herniated intervertebral disc, compressing the L5 or S1 ventral rami. The so-called piriformis syndrome, which may result from a variant or anomalous relationship between piriformis and the sciatic nerve, may also produce hip pain that radiates inferiorly along the course of the sciatic nerve.

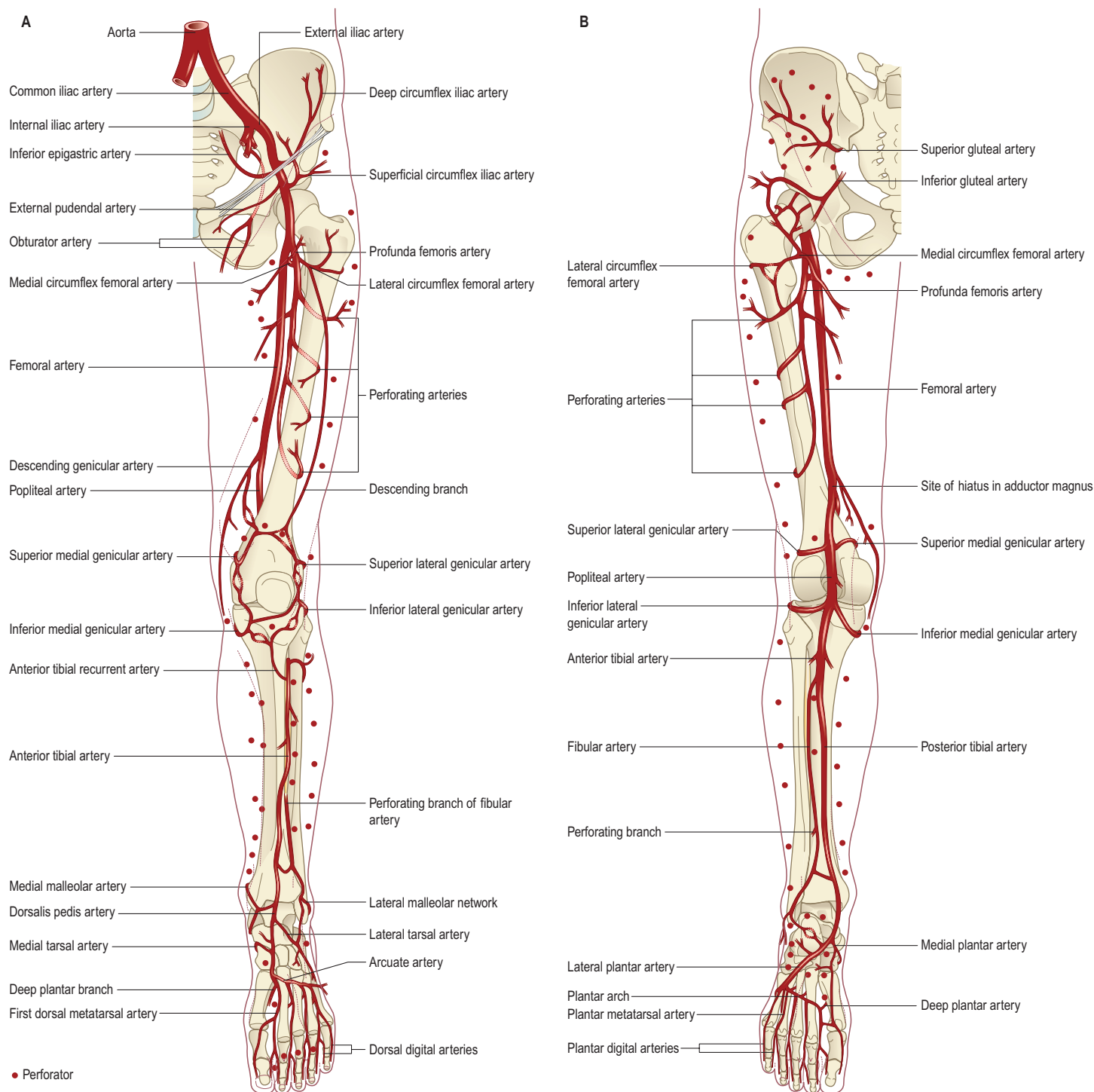


Fig. 78.7 The source and location of the arterial perforators of the lower limb. **A**, Anterior aspect. **B**, Posterior aspect.

OVERVIEW OF THE PRINCIPAL NERVES OF THE LOWER LIMB

Femoral nerve (L2–4)

The femoral nerve is the nerve of the anterior compartment of the thigh. It arises from the posterior divisions of the second to fourth lumbar ventral rami, descends through psoas major and emerges on its lateral border to pass between it and iliacus. It enters the thigh behind the inguinal ligament and lateral to the femoral sheath. Its terminal branches form in the femoral triangle about 2 cm distal to the inguinal ligament. In the abdomen, the nerve supplies small branches to iliacus and a branch to the proximal part of the femoral artery. It subsequently supplies a large cutaneous area on the anterior and medial thigh, medial leg and foot, and gives articular branches to the hip, knee and ankle. The femoral nerve is described in detail on [page 1372](#).

Obturator nerve (L2–4)

The obturator nerve is the nerve of the medial compartment of the thigh. It arises from the anterior divisions of the second to fourth lumbar ventral rami, descends through psoas major and emerges from its medial border at the pelvic brim. It crosses the sacroiliac joint behind the common iliac artery and lateral to the internal iliac vessels, runs along the lateral pelvic wall medial to obturator internus, and enters the thigh through the upper part of the obturator foramen. Near the foramen, it divides into anterior and posterior branches, which are separated at first by part of obturator externus and more distally by adductor brevis. It provides articular branches to the hip and knee, and may supply skin on the medial thigh and leg. The obturator nerve is described in detail on [page 1372](#). An accessory obturator nerve may also be present and leaves the pelvis anterior to the pubis.

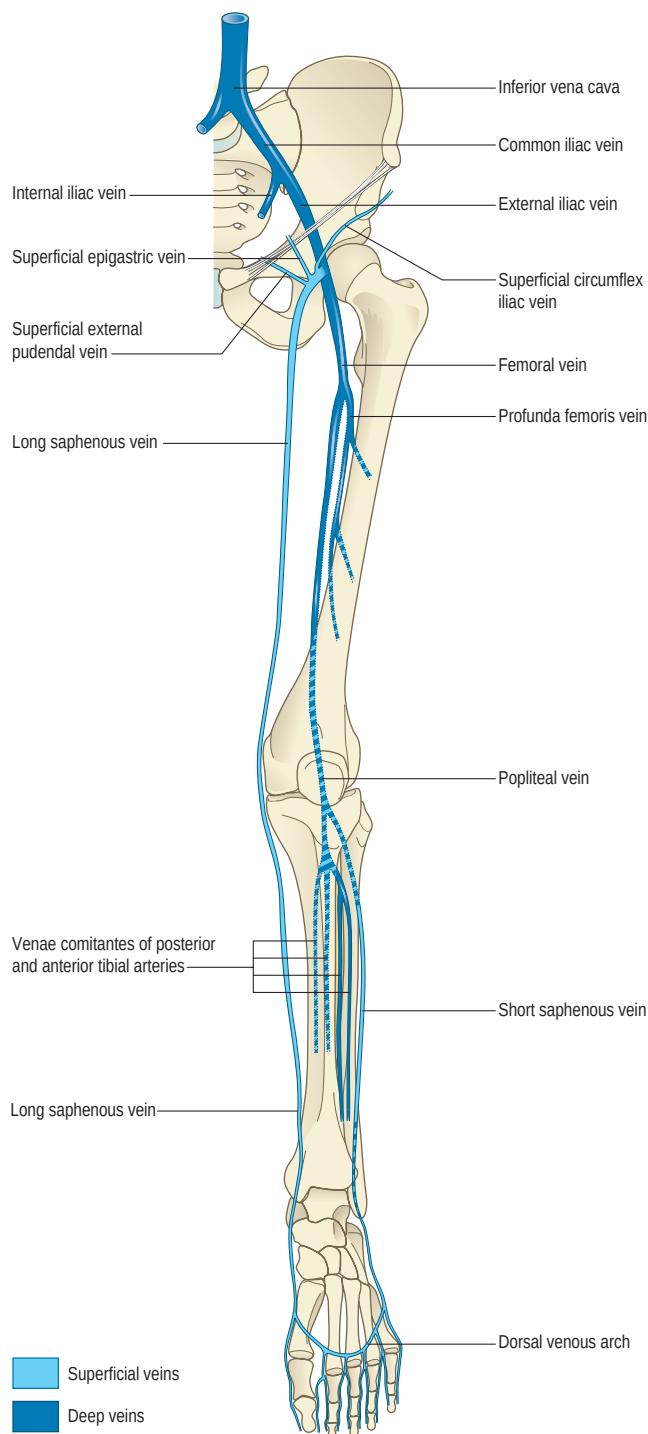


Fig. 78.8 An overview of the veins of the lower limb.

Sciatic nerve (L4, L5, S1–3)

The sciatic nerve travels in the posterior compartment of the thigh and, via its major branches, supplies the leg and foot. Formed in the pelvis from the ventral rami of the fourth lumbar to third sacral spinal nerves, it is typically 2 cm wide at its origin and is the thickest nerve in the body. It enters the lower limb via the greater sciatic foramen inferior to piriformis and descends between the greater trochanter and ischial tuberosity. In its descent along the posterior thigh, it is crossed by the long head of biceps femoris and divides into the tibial and common fibular (peroneal) nerves proximal to the knee; however, the level of bifurcation can be variable. Prior to diverging, the tibial and common fibular nerves are structurally separate and only loosely held together as the sciatic nerve. The tibial nerve is derived from the anterior divisions of the sacral plexus, and the common fibular nerve is made up of the posterior divisions of the plexus. The sciatic nerve sends articular branches to the hip joint through its posterior capsule (these are sometimes derived directly from the sacral plexus) and to the knee joint. The posterior thigh muscles, including the ischial part of adductor magnus

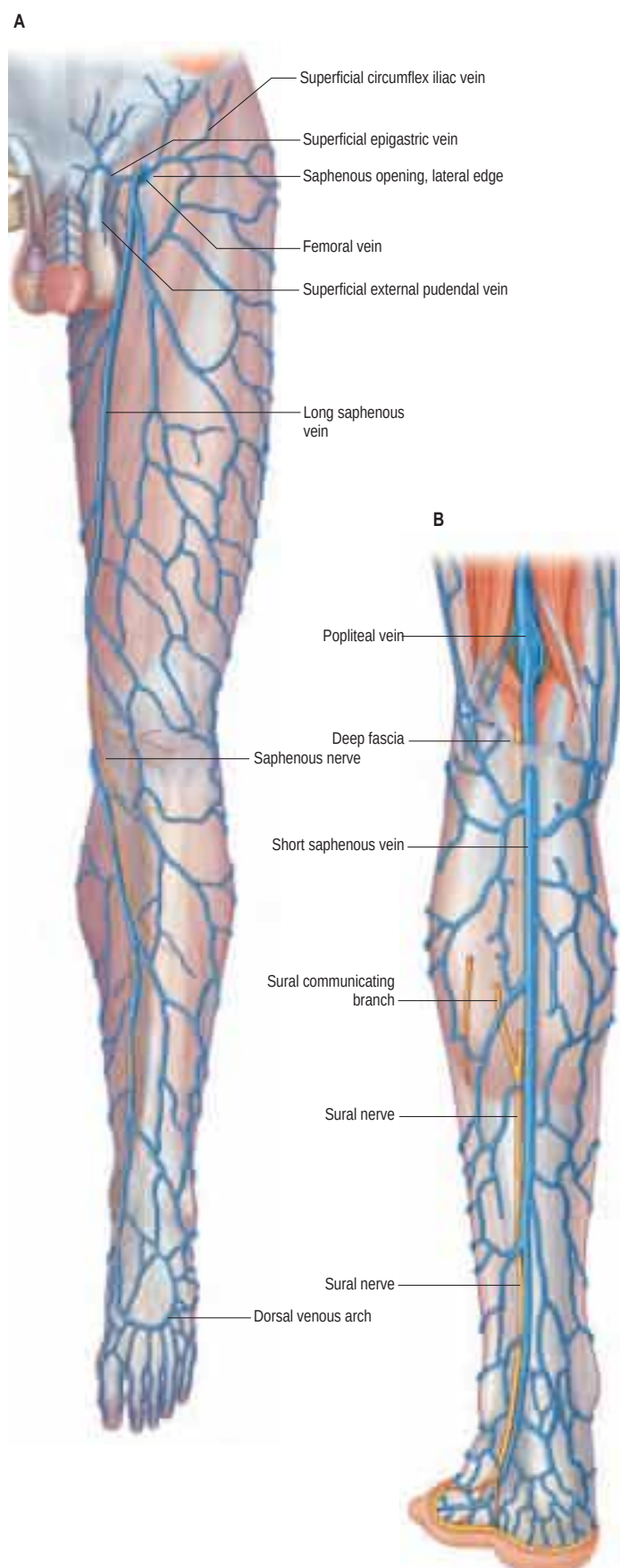


Fig. 78.9 **A**, The long saphenous vein and its tributaries. **B**, The short saphenous vein and its tributaries.

but not the short head of biceps femoris, are supplied by the medial (tibial) component of the sciatic nerve. The short head of biceps femoris is supplied by the lateral (common fibular) component. The common variations of the sciatic nerve have been well described and have been classified into six types (Beason and Anson 1937). The sciatic nerve is described in detail on [page 1373](#).

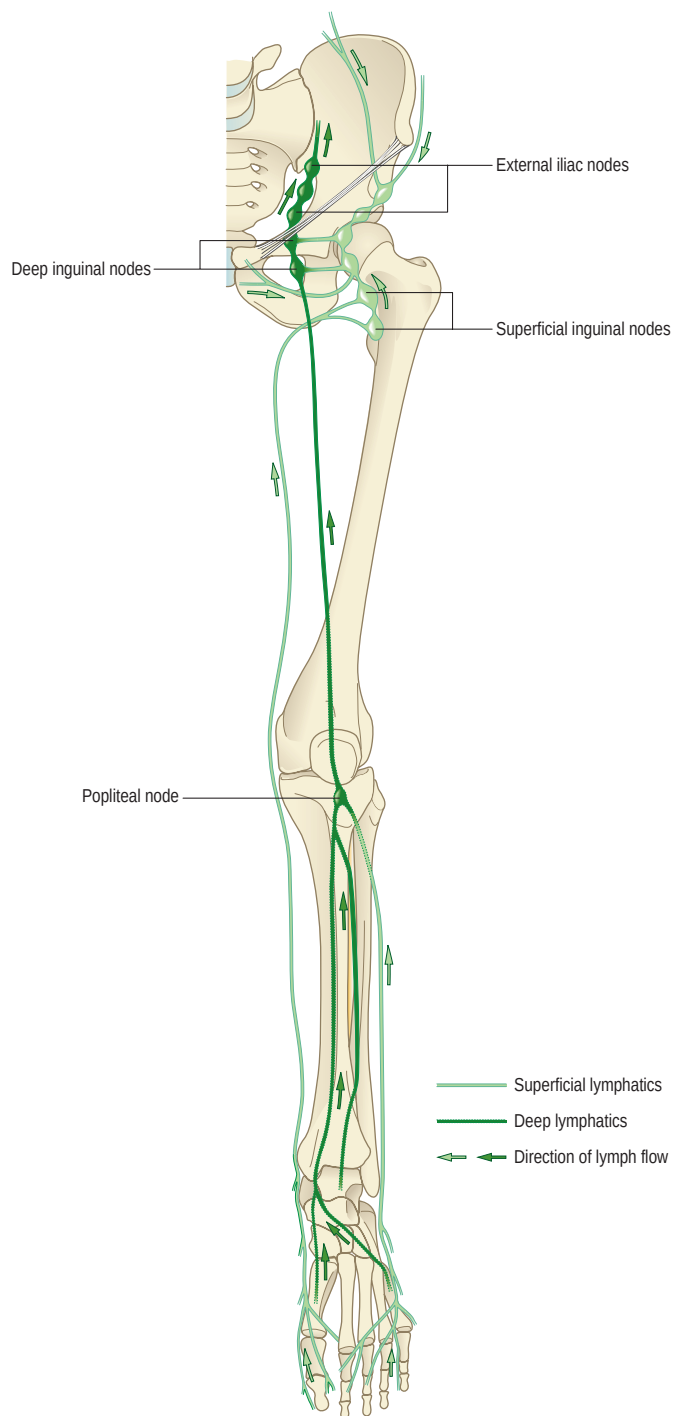


Fig. 78.10 An overview of the lymphatics of the lower limb.

Tibial nerve (L4, L5, S1–3)

The tibial nerve is derived from the anterior divisions of the sacral plexus. It descends along the posterior thigh and popliteal fossa to the distal border of popliteus, then passes anterior to the soleus with the popliteal artery and continues into the leg. In the popliteal fossa, it lies lateral to the popliteal vessels, becomes superficial to them at the knee and crosses to the medial side of the artery. In the leg, it is the nerve of the posterior compartment and descends with the posterior tibial vessels to lie between the heel and the medial malleolus. It ends deep to the flexor retinaculum by dividing into the medial and lateral plantar nerves. The tibial nerve supplies articular branches to the knee and ankle joints. Its cutaneous supply, including its terminal branches, supplies the back of the calf, the sole, the lateral border of the foot and the medial and lateral sides of the heel. The tibial nerve is described in detail on [page 1415](#).

Common fibular nerve (L4, L5, S1, S2)

The common fibular nerve (common peroneal nerve) is derived from the posterior divisions of the sacral plexus. In the leg, it is the nerve of

the anterior and lateral compartments. It descends obliquely along the lateral side of the popliteal fossa to the fibular head, lying between the tendon of biceps femoris and the lateral head of gastrocnemius; it then curves lateral to the neck of the fibula, lying on the bone deep to fibularis longus, and divides into superficial and deep fibular (peroneal) nerves. Before it divides, it gives branches to the knee and superior tibiofibular joints, and to the skin. The cutaneous area supplied by the common fibular nerve and its terminal branches includes the antero-lateral leg and most of the dorsum of the foot. The common fibular nerve is described in detail on [page 1415](#).

Gluteal nerves (L4, L5, S1, S2)

The gluteal nerves arise from the posterior divisions of the sacral plexus. The superior gluteal nerve (L4, L5, S1) leaves the pelvis through the greater sciatic foramen superior to piriformis and supplies gluteus medius, gluteus minimus, tensor fasciae latae and the hip joint. The inferior gluteal nerve (L5, S1, S2) passes through the greater sciatic foramen inferior to piriformis and supplies gluteus maximus. The gluteal nerves are described in detail on [page 1374](#).

CUTANEOUS INNERVATION

The cutaneous nerves supplying the skin of the lower limb are all branches of the lumbar and sacral plexuses, with the exception of some proximal nerves. The areas of distribution and spinal segments of origin of the cutaneous nerves of the lower limb are illustrated in [Figure 78.12](#). Variations in the composition and course of the cutaneous nerves of the lower limb are common.

Dermatomes

Our knowledge of the extent of individual dermatomes, especially in the limbs, is largely based on clinical evidence ([Ladak et al 2014](#)). The dermatomes of the lower limb are innervated by spinal nerves T12 to S3 (see [Fig. 78.12](#); [Figs 78.13–78.14](#)).

AUTONOMIC INNERVATION

The autonomic nerve supply to the limbs is exclusively sympathetic. Preganglionic sympathetic fibres to the lower limb are derived from neurones in the lateral horn of the lower thoracic (T10, T11 and T12) and upper lumbar (L1, L2) spinal cord segments. Fibres pass in white rami communicantes to the sympathetic trunk and synapse in the lumbar and sacral ganglia. Postganglionic fibres pass in grey rami communicantes to enter the lumbar and sacral plexuses; many are distributed to the skin via the cutaneous branches of the nerves derived from these plexuses. The blood vessels to the lower limb receive their sympathetic nerve supply via adjacent peripheral nerves. Postganglionic fibres accompanying the iliac arteries are destined mainly for the pelvis but may supply vessels in the proximal thigh.

MOVEMENTS, MUSCLES AND SEGMENTAL INNERVATION

Most limb muscles are innervated by neurones derived from more than one segment of the spinal cord. The predominant segmental origin of the nerve supply for each of the muscles of the lower limb and for the movements that take place at the joints of the lower limb is summarized in [Tables 78.1–78.4](#) ([Sharrard 1955](#)).

Movements At the central nervous level of control, muscles are not recognized as individual actuators but as components of movement, and may therefore contribute to several types of motion, acting variously as prime movers, antagonists, fixators or synergists. Some muscles have been included in more than one place in [Table 78.3](#) on the basis that a muscle that acts across one joint can produce a combination of movements (e.g. flexion with medial rotation, or extension with adduction) and a muscle that crosses two joints can produce more than one movement. It is also important to remember that these listings are not exhaustive.

Spinal nerves There is no universal consensus concerning the contribution that individual spinal nerves make to the innervation of individual muscles; the most positive identifications, which are limited,

For example, the ilioinguinal and iliohypogastric nerves may arise from a common trunk or the ilioinguinal nerve may be absent. The ilioinguinal nerve may also join the iliohypogastric nerve at the iliac crest. When the obturator nerve makes a more significant contribution to the cutaneous innervation, the medial cutaneous branch of the femoral nerve is relatively small. Alternatively, the cutaneous branch of the obturator nerve may be absent. The lateral femoral cutaneous nerve normally arises from L2 and L3, but L1 may also contribute. Although it usually bifurcates after it exits the pelvis, it may bifurcate within the pelvic cavity. The nerve may be absent on one side and/or may be replaced by the ilioinguinal nerve or a branch of the anterior femoral cutaneous nerve. The genital and femoral branches of the genitofemoral nerve may arise as separate offshoots of the lumbar plexus. The genital branch may receive fibres from the twelfth thoracic nerve or may be completely absent, while the femoral branch may have an extensive distribution to the skin of the upper two-thirds of the thigh. The sural nerve is subject to wide variation and may supply the dorsal cutaneous aspect of the lateral two-and-a-half toes, or may terminate in the foot without any digital branches.

The preaxial border starts near the midpoint of the thigh and descends to the knee. It then curves medially, descending to the medial

malleolus and the medial side of the foot and hallux. The postaxial border starts in the gluteal region and descends to the centre of the popliteal fossa, then deviates laterally to the lateral malleolus and the lateral side of the foot. The ventral and dorsal axial lines exhibit corresponding obliquity. The ventral axial line starts proximally at the medial end of the inguinal ligament and descends along the postero-medial aspect of the thigh and leg to end proximal to the heel. The dorsal axial line begins in the lateral gluteal region and descends posterolaterally in the thigh to the knee; it inclines medially and ends proximal to the ankle. Considerable overlap exists between adjacent dermatomes innervated by nerves derived from consecutive spinal cord segments.

Surgical or chemical lumbar sympathectomy may be indicated in arterial disease and in the management of plantar hyperhidrosis, and may be used to treat rest pain or other troublesome sensory symptoms of arterial disease or in causalgia. A segment of the sympathetic trunk including the second and third lumbar ganglia is removed; preservation of the first lumbar ganglion is said to lessen the risk of ejaculatory problems.

Table 78.1 Segmental innervation of the muscles of the lower limb

Segment	Muscles supplied
L1	Psoas major, psoas minor
L2	Psoas major, iliacus, sartorius, gracilis, pectineus, adductor longus, adductor brevis
L3	Psoas major, quadriceps femoris, adductors (magnus, longus, brevis)
L4	Psoas major, quadriceps femoris, tensor fasciae latae, adductor magnus, obturator externus, tibialis anterior, tibialis posterior
L5	Gluteus medius, gluteus minimus, obturator internus, semimembranosus, semitendinosus, extensor hallucis longus, extensor digitorum longus, fibularis tertius, popliteus
S1	Gluteus maximus, obturator internus, piriformis, biceps femoris, semitendinosus, popliteus, gastrocnemius, soleus, fibularis longus and fibularis brevis, extensor digitorum brevis
S2	Piriformis, biceps femoris, gastrocnemius, soleus, flexor digitorum longus, flexor hallucis longus, some intrinsic foot muscles
S3	Some intrinsic foot muscles (except abductor hallucis, flexor hallucis brevis, flexor digitorum brevis, extensor digitorum brevis)

Table 78.2 Segmental innervation of joint movements of the lower limb

Region	Muscles supplied	Segment
Hip	Flexors, adductors, medial rotators	L1–3
	Extensors, abductors, lateral rotators	L5, S1
Knee	Extensors	L3, 4
	Flexors	L5, S1
Ankle	Dorsiflexors	L4, 5
	Plantar flexors	S1, 2
Foot	Invertors	L4, 5
	Evertors	L5, S1
	Intrinsic muscles	S2, 3

PELVIC GIRDLE AND LOWER LIMB: OVERVIEW AND SURFACE ANATOMY

Table 78.3 Movements, muscles and segmental innervation in the lower limb*

Joint	Movement	Muscle	Innervation	L1	L2	L3	L4	L5	S1	S2	S3
HIP	FLEXION	Psoas major	Spinal nn. L1–3								
		Iliacus	Femoral n.								
		Pectineus	Femoral n. or accessory obturator n.								
		Rectus femoris	Femoral n.								
		Adductor longus	Obturator n.								
		Sartorius	Femoral n.								
	EXTENSION	Gluteus maximus	Inferior gluteal n.								
		Adductor magnus	Obturator and tibial nn.								
		Semitendinosus, semimembranosus, biceps femoris	Tibial and common fibular nn.								
	MEDIAL ROTATION	Iliacus	Femoral n.								
		Gluteus medius and minimus	Superior gluteal n.								
		Tensor fasciae latae	Superior gluteal n.								
	LATERAL ROTATION	Superior and inferior gemelli	Nerve to obturator internus and nerve to quadratus femoris, respectively								
		Quadratus femoris	Nerve to quadratus femoris								
		Piriformis	Nerve to piriformis								
		Obturator internus	Nerve to obturator internus								
		Obturator externus	Obturator n.								
		Sartorius	Femoral n.								
	ADDUCTION	Gracilis	Obturator n.								
		Adductor longus	Obturator n.								
		Adductor magnus	Obturator and tibial nn.								
		Adductor brevis	Obturator n.								
		Pectineus	Femoral n. or accessory obturator n.								
	ABDUCTION	Tensor fasciae latae	Superior gluteal n.								
		Gluteus medius and minimus	Superior gluteal n.								
		Piriformis	Nerve to piriformis								
KNEE	FLEXION	Semimembranosus	Tibial n.								
		Semitendinosus	Tibial n.								
		Biceps femoris	Tibial and common fibular nn.								
		Gastrocnemius	Tibial n.								
	EXTENSION	Quadriceps femoris:									
		Rectus femoris	Femoral n.								
		Vastus lateralis	Femoral n.								
		Vastus intermedius	Femoral n.								
		Vastus medialis	Femoral n.								
ANKLE	DORSIFLEXION	Tibialis anterior	Deep fibular n.								
		Extensor digitorum longus	Deep fibular n.								
		Extensor hallucis longus	Deep fibular n.								
		Fibularis tertius	Deep fibular n.								
	PLANTAR FLEXION	Gastrocnemius	Tibial n.								
		Soleus	Tibial n.								
		Flexor digitorum longus	Tibial n.								
		Flexor hallucis longus	Tibial n.								
		Fibularis longus	Superficial fibular n.								
		Tibialis posterior	Tibial n.								
	INVERSION	Tibialis anterior	Deep fibular n.								
		Tibialis posterior	Tibial n.								
	EVERSION	Fibularis longus	Superficial fibular n.								
		Fibularis tertius	Deep fibular n.								
		Fibularis brevis	Superficial fibular n.								
TOES	FLEXION	Flexor digitorum longus	Tibial n.								
		Flexor hallucis longus	Tibial n.								
		Flexor hallucis brevis	Medial plantar n.								
		Flexor digitorum brevis	Medial plantar n.								
		Flexor accessorius (quadratus plantae)	Lateral plantar n.								
		Flexor digiti minimi brevis	Lateral plantar n.								
		Abductor hallucis	Medial plantar n.								
		Abductor digiti minimi	Lateral plantar n.								
		Lumbricals	Medial and lateral plantar nn.								
	EXTENSION	Extensor digitorum longus	Deep fibular n.								
		Extensor hallucis longus	Deep fibular n.								
		Extensor digitorum brevis	Deep fibular n.								
	ABDUCTION	Abductor hallucis	Medial plantar n.								
		Abductor digiti minimi	Lateral plantar n.								
		Dorsal interossei	Lateral plantar n.								
	ADDUCTION	Plantar interossei	Lateral plantar n.								
		Adductor hallucis	Lateral plantar n.								

*Royal blue shading denotes nerve roots from which there is a known dominant contribution. Turquoise shading denotes nerve roots from which the contribution is of similar degree.

Table 78.4 The movements and muscles tested to determine the location of a lesion in the lower limb

Movement	Muscle	Upper motor neurone*	Spinal nerve level	Reflex	Nerve
Hip flexion	Iliopsoas	++	L1, 2		Femoral
Hip adduction	Adductors	+	L2, 3	(+)	Obturator
Hip extension	Gluteus maximus		L5, S1		Inferior gluteal
Knee flexion	Hamstrings	+	S1		Sciatic
Knee extension	Quadriceps femoris		L3, 4	++	Femoral
Ankle dorsiflexion	Tibialis anterior	++	L4		Deep fibular
Ankle eversion	Fibularis longus and fibularis brevis		L5, S1		Superficial fibular
Ankle inversion	Tibialis posterior		L4, 5		Tibial
Ankle plantar flexion	Gastrocnemius/soleus	+	S1, 2	++	Tibial
Great toe extension	Extensor hallucis longus		L5		Deep fibular

*The muscles listed in the 'Upper motor neurone' column are those that are preferentially affected in upper motor neurone lesions. The root level is the principal supply to a muscle.

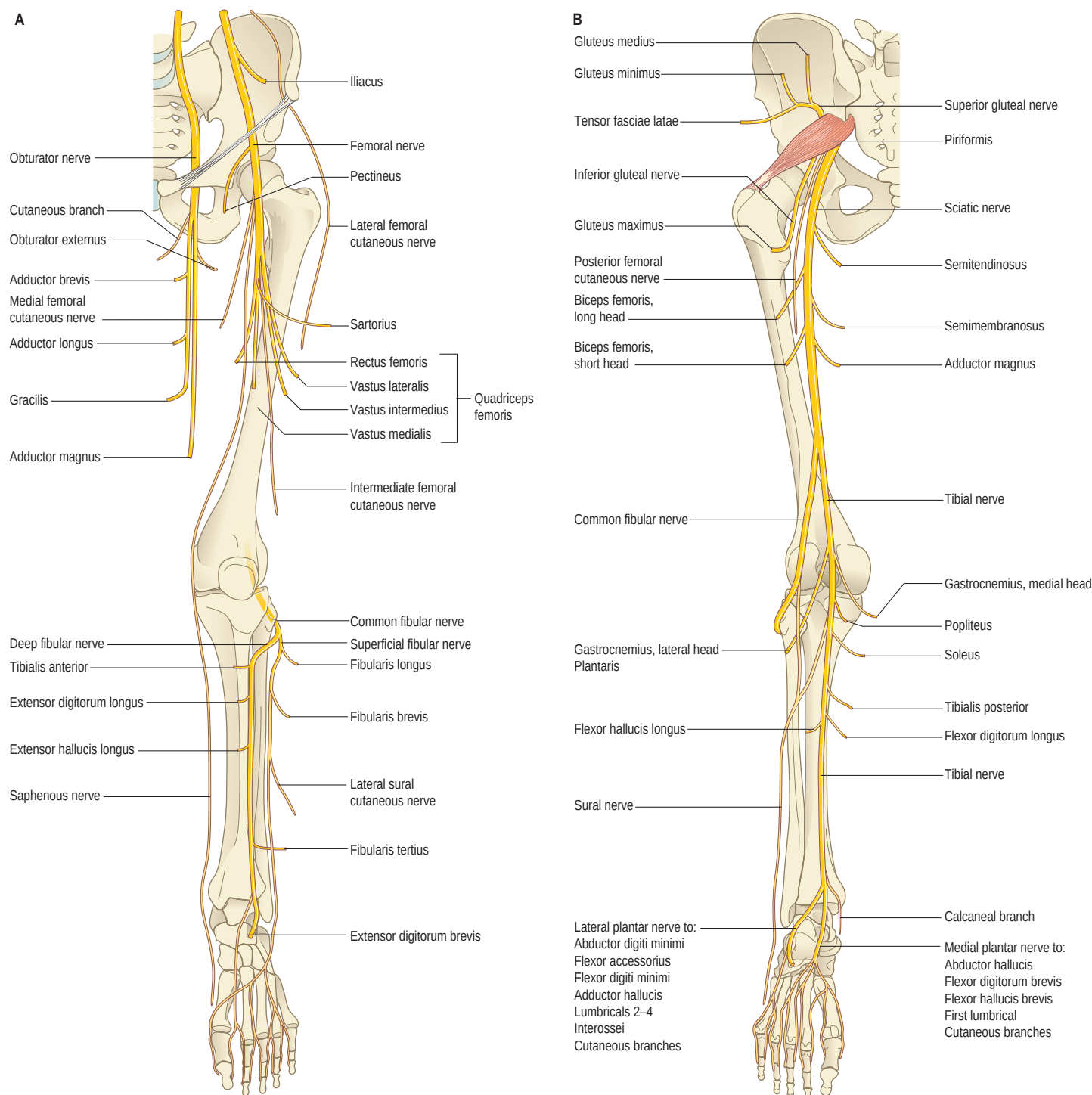


Fig. 78.11 The nerves of the lower limb, with their cutaneous and muscular branches. **A**, Anterior aspect. **B**, Posterior aspect. Adductor magnus has dual innervation.



have been obtained by electrically stimulating spinal nerves and recording the evoked electromyographic activity in the muscles. Much of the information in [Tables 78.1–78.4](#) is based on the clinical appreciation of deficits following lesions to various nerves.

Reflexes

Knee reflex (L2–4) With the patient sitting and the knee supported and partially flexed, the patellar ligament is struck with a finger or percussion hammer, resulting in extension of the knee joint.

Ankle reflex (S1, 2) With the patient sitting and the lower limb laterally rotated and partially flexed at the hip and knee, the foot is dorsiflexed by the examiner and the calcaneal tendon struck with a percussion hammer. This results in plantar flexion of the foot.

Plantar reflex The plantar reflex is an important part of the clinical examination of the central nervous system. With the foot relaxed, the outer edge of the sole is stroked longitudinally with a blunt object such

as the tip of the handle of a percussion hammer. Normally, this action elicits flexion of the toes. However, in patients with upper motor neurone lesions, the response includes extension of the great toe (Babinski's sign).

Neurological localization of a lesion



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SURFACE ANATOMY

SKELETAL LANDMARKS

Pelvis

The inguinal or groin skin crease marks the junction of the anterior thigh with the anterior abdominal wall. It lies distal to the inguinal

In clinical practice, it is only necessary to test a relatively small number of muscles in order to determine the location of a lesion. Any muscle to be tested must satisfy a number of criteria. It should be visible, so that wasting or fasciculation can be observed, and the muscle consistency with contraction can be felt. It should have an isolated action, so that its function can be tested separately. It should help to differentiate between lesions at different levels in the neural axis and in the peripheral nerve, or between peripheral nerves. It should be tested in such a way that normal can be differentiated from abnormal, so that slight weakness can be detected early with reliability. Some preference should be given to muscles with an easily elicited reflex. [Table 78.4](#) gives a list of movements and muscles chosen according to these criteria; in practice, these tests would be combined with tests of sensory function. Knowledge of the sequence in which motor branches leave a peripheral nerve to innervate specific muscles is very helpful in localizing the level of a lesion.

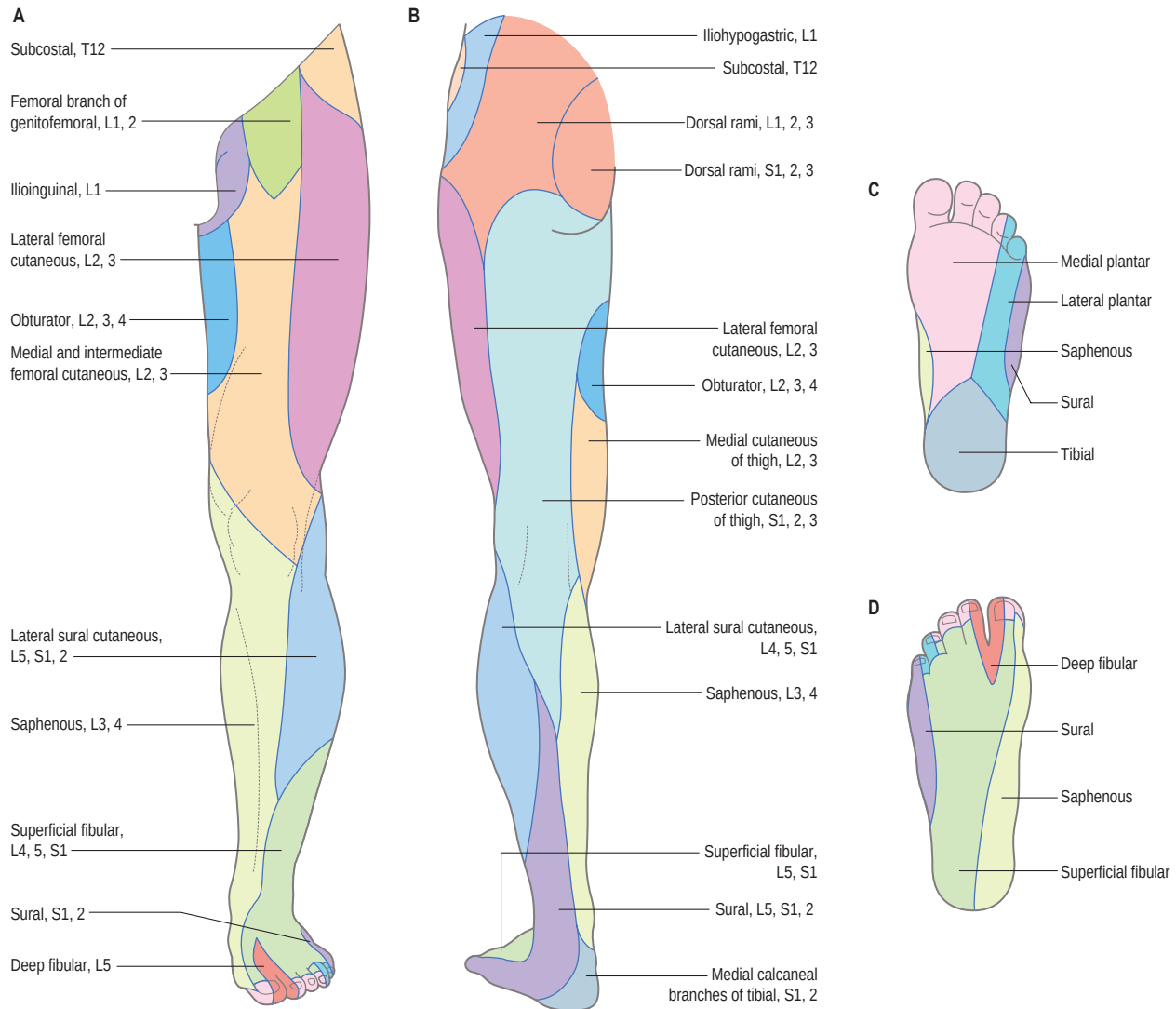
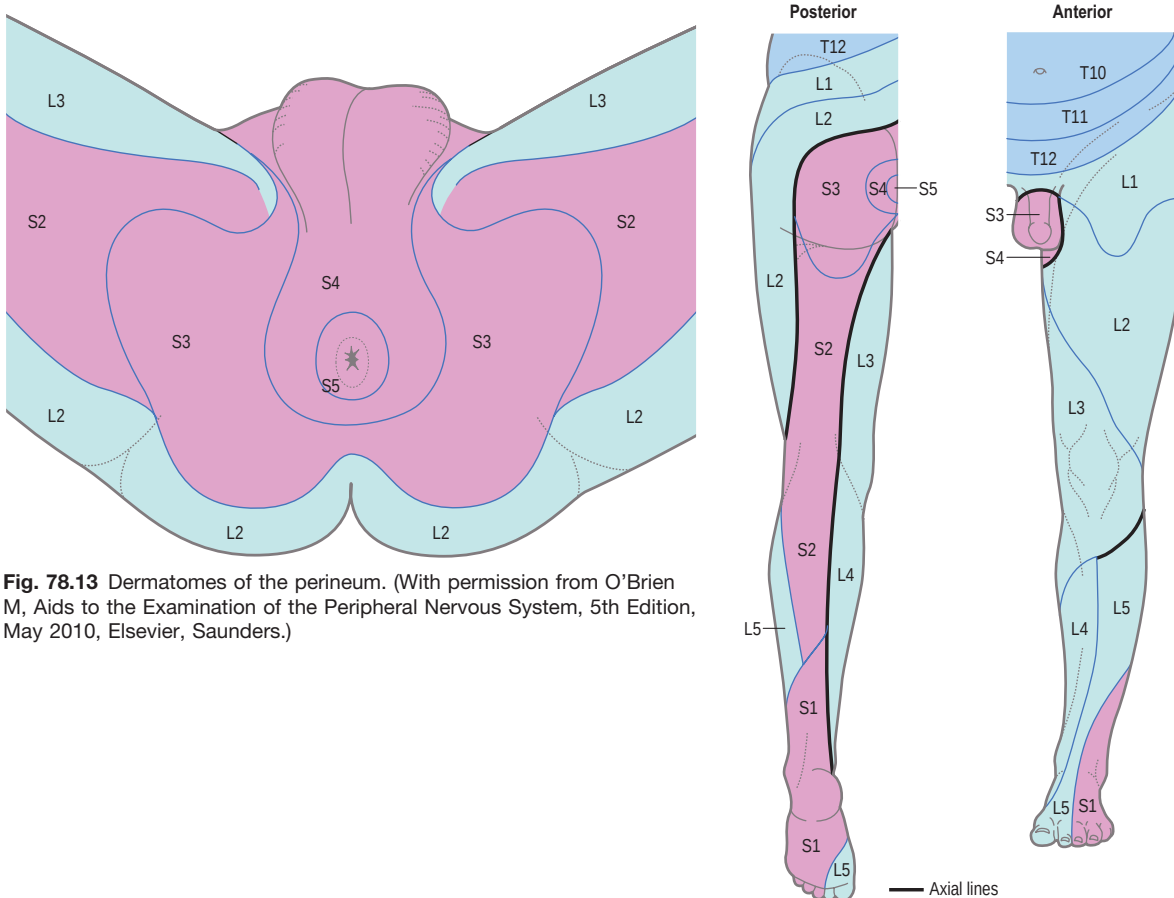


Fig. 78.12 The cutaneous nerves of the lower limb: areas of distribution and spinal segments of origin. **A**, Anterior aspect. **B**, Posterior aspect. **C**, The sole of the foot. **D**, The dorsum of the foot.



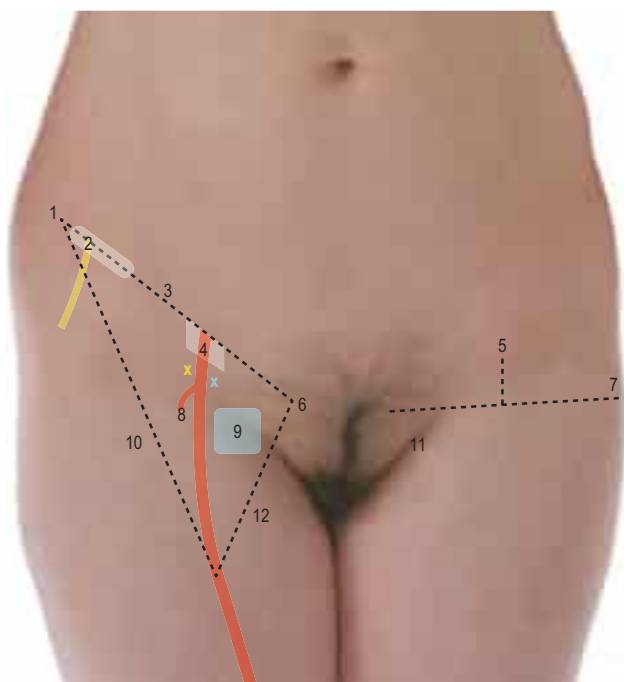


Fig. 78.15 The inguinal region and femoral triangle. Key: 1, anterior superior iliac spine; 2, lateral femoral cutaneous nerve and its zone of emergence into femoral triangle (white): range 0.3–7.3 cm (mean 2.1–3.5 cm) from anterior superior iliac spine; 3, inguinal ligament; 4, femoral artery and zone of emergence into femoral triangle (white): mid-inguinal point ± 1 cm either side, femoral nerve (yellow cross) and vein (blue cross) sit lateral and medial, respectively, to the femoral artery; 5, femoral head: located 2–4 cm above the midpoint of the greater trochanter to pubic tubercle line; 6, pubic tubercle; 7, greater trochanter to pubic tubercle line; 8, profunda femoris artery: almost always arises 6.5 $\pm$ 1.5 cm distal to the mid-inguinal point, or superior to or at the level of the inguinal/groin crease; 9, zone of saphenofemoral junction location: almost always sits within a 3 cm $\times$ 3 cm zone situated 1–4 cm lateral and 0–3 cm inferior to the pubic tubercle; 10, sartorius; 11, inguinal/groin crease; 12, adductor longus. (Adapted from Drake R, Vogl AW, Mitchell AW, et al, Gray's Atlas of Anatomy, Elsevier 2008, Churchill Livingstone.)

ligament by a mean of 6.7 $\pm$ 1.9 cm (Lechner et al 1988) (Fig. 78.15). The inguinal crease sits more distal to the inguinal ligament in females than in males (7.5 $\pm$ 1.9 cm versus 6.3 $\pm$ 1.9 cm) (Lechner et al 1988). The anterior superior iliac spine lies superior to the lateral end of the fold, and from this point, the iliac crest is palpable along its entire length. A line (Tuffier's) joining the most superior aspects of the iliac crests almost always crosses the L4 body or L4/L5 intervertebral disc (Render 1996, Walsh et al 2006, Chakraverty et al 2007) (Fig. 78.16). This line is used as a landmark when performing a lumbar puncture; localization via palpation, especially in females and patients with a greater body mass index (BMI), often places it at a higher vertebral level, up to the L2–L3 interspace (Chakraverty et al 2007, Kim et al 2007, Hale et al 2010). Vertebral column flexion in the adult has little effect on the level of Tuffier's line (Kim et al 2003). In neonates, Tuffier's line sits at the L4/L5 interlaminar space level whilst prone, moving to the upper third of L5 during vertebral column flexion (van Schoor et al 2014). The iliac crest terminates posteriorly as the posterior superior iliac spine, which is marked superficially by the gluteal dimples (dimples of Venus). Palpation places the posterior superior iliac spines in the range of the L5–S1 vertebral junction to the S2 spinous process level (Kim et al 2007), whereas radiographic assessment shows their inferior margin to mark vertebral level S2 almost always (McGaugh et al 2007). The caudal limit of the dura mater ranges from the L5–S1 junction to S4 (Hansasuta et al 1999, Binokay et al 2006, Senoglu et al 2013), with the majority sitting at the lower S1 to the S2 level (Senoglu et al 2013) and the mean position being lower in males versus females (Binokay et al 2006). The ischial tuberosity is covered by gluteus maximus and is palpable during hip flexion. The coccyx is palpable in the superior part of the intergluteal cleft, the deep vertical groove between the buttocks extending to the S3–S4 bony sacral segments. The sacral hiatus sits superior to the tip of the coccyx at a mean of 57.5 $\pm$ 8.7 mm and is flanked by the raised palpable sacral cornua, which serve as reliable landmarks (Aggarwal et al 2009). The hiatus can also be located at the apex of an inverted equilateral triangle

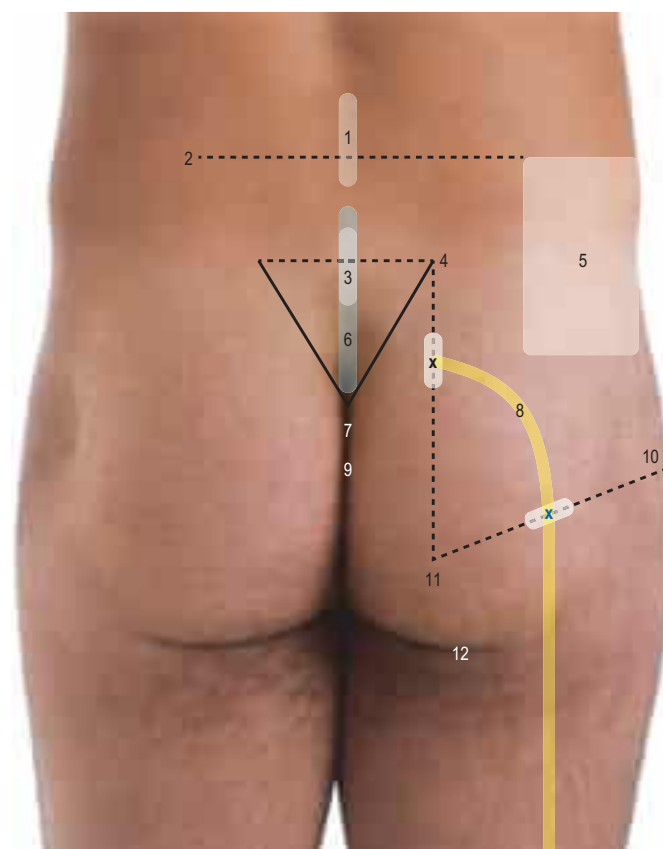


Fig. 78.16 The gluteal region and sciatic nerve. Key: 1, zone of supracristal plane intersection with vertebrae (white): ranges from the L2–L3 to the L4–L5 junction/interspinous space; 2, highest point of iliac crests (supracristal plane or Tuffier's line): most commonly intersects the L4 vertebra to the L4/5 junction; 3, zone of intersection between the vertebrae and a horizontal plane between the posterior superior iliac spines (white): ranges from the L5–S1 junction/interspinous space to S2 when determined via palpation, located at S2 via radiographs; 4, posterior superior iliac spine and interspinous plane: marked by a skin indentation (dimple of Venus); 5, safe zone (white) for gluteal region injection (upper lateral quadrant), determined via a vertical line from the highest point of the iliac crest to ischial tuberosity level, and a horizontal line passing through its midpoint; 6, zone of termination of dural sac (blue): ranges from L5–S1 to S4, most commonly sitting from the lower part of S1 to the S2 level; 7, sacral hiatus bordered by sacral cornua, located at the apex of an approximately equilateral triangle with baseline measurement taken between the posterior superior iliac spines; 8, course of sciatic nerve: from approximately one third of the way down a line joining the posterior superior iliac spine to the ischial tuberosity (black X and white zone) to the midpoint or middle third of a line joining the ischial tuberosity to the upper greater trochanter (blue X and white zone); 9, tip of coccyx within intergluteal cleft; 10, greater trochanter; 11, ischial tuberosity; 12, gluteal fold. (Adapted from Drake R, Vogl AW, Mitchell AW, et al, Gray's Atlas of Anatomy, Elsevier 2008, Churchill Livingstone.)

whose base is measured between the posterior superior iliac spines (Senoglu et al 2005), although this arrangement may not always be observed (Aggarwal et al 2009). The termination of the dural sac is located a mean of 31.6 $\pm$ 11.8 mm from the sacral hiatus apex (Aggarwal et al 2009), a point relevant to caudal anaesthetic administration.

Femur

The greater trochanter of the femur sits on the upper lateral thigh, inferior to the midpoint of the iliac crest and approximately level with a horizontal plane passing through the pubic tubercle (see Fig. 78.15). It can be used to guide surgeons during anterolateral approaches to the hip joint. A line (Nelaton's/Roser–Nelaton's) drawn from the anterior superior iliac spine to the ischial tuberosity passes superior to or over the greater trochanter. In a patient with hip dislocation or femoral neck fracture, the greater trochanter can be palpated superior to this line. The femoral shaft is not directly palpable but runs inferomedially from a point 2–3 cm medial to the lateral aspect of the greater trochanter to the midpoint of the patella. The femoral head sits 2–4 cm superior to the midpoint of a line between the superior margin of the greater trochanter and the pubic tubercle.

When the knee is flexed, the medial surface of the medial condyle and the lateral surface of the lateral condyle of the femur may both be palpated, and portions of the femoral condylar articular surface can be examined on each side of the patellar ligament.

Patella

The entire patella is palpable except for its articular surface. When the quadriceps femoris is relaxed in the fully extended knee, the patella can be tilted and moved on the distal aspect of the femur. The inferior limit of the patella is located superior to the knee joint line by 1–2 cm (Fig. 78.17).

Knee

The patella and the femoral condyles have been described above. The tibial condyles form visible and palpable landmarks on the medial and lateral sides of the patellar ligament (see Fig. 78.17). The latter may be traced inferiorly from the apex of the patella to the tibial tuberosity, which is both visible and palpable. When the knee is flexed, the anterior margins of the tibial condyles can be felt: each forms the lower boundary of a depression at the side of the patellar ligament. The lateral condyle is the more prominent of the two. The joint line of the knee (tibiofemoral joint) corresponds to the upper margins of the tibial condyles and can be represented by a line drawn round the limb at this level. The anterior horns of the menisci lie in the angles between this

line and the edges of the patellar ligament. The iliotibial tract is attached to a prominence, Gerdy's tubercle, on the anterior aspect of the lateral condyle that is usually approximately 1 cm inferior to the joint line and about 2 cm superolateral to the tibial tuberosity.

The head of the fibula forms a slight surface elevation on the posterior part of the proximal lateral leg. It lies vertically below the posterior part of the lateral condyle of the femur, 1–2 cm inferior to the knee joint line.

Leg and ankle

The subcutaneous medial surface of the tibia corresponds to the flat anteromedial aspect of the leg. Superiorly, this surface merges with the medial condyle of the tibia, and inferiorly, it is continuous with the visible prominence of the medial malleolus of the tibia (see Fig. 78.17; Figs 78.18–78.19). The anterior border of the tibia can be palpated distinctly throughout most of its extent, but inferiorly, it is somewhat masked by the tendon of tibialis anterior, which lies to its lateral side. The lateral malleolus of the fibula forms a conspicuous projection on the lateral side of the ankle; it descends to a more distal level than the medial malleolus and is located on a more posterior plane. The lateral aspect of the lateral malleolus is continuous superiorly with an elongated, subcutaneous, triangular area of the inferior shaft of the fibula. The lateral part of the anterior margin of the inferior end of the tibia can be palpated immediately anterior to the base of the lateral malleolus; the line of the ankle joint can be gauged from it.

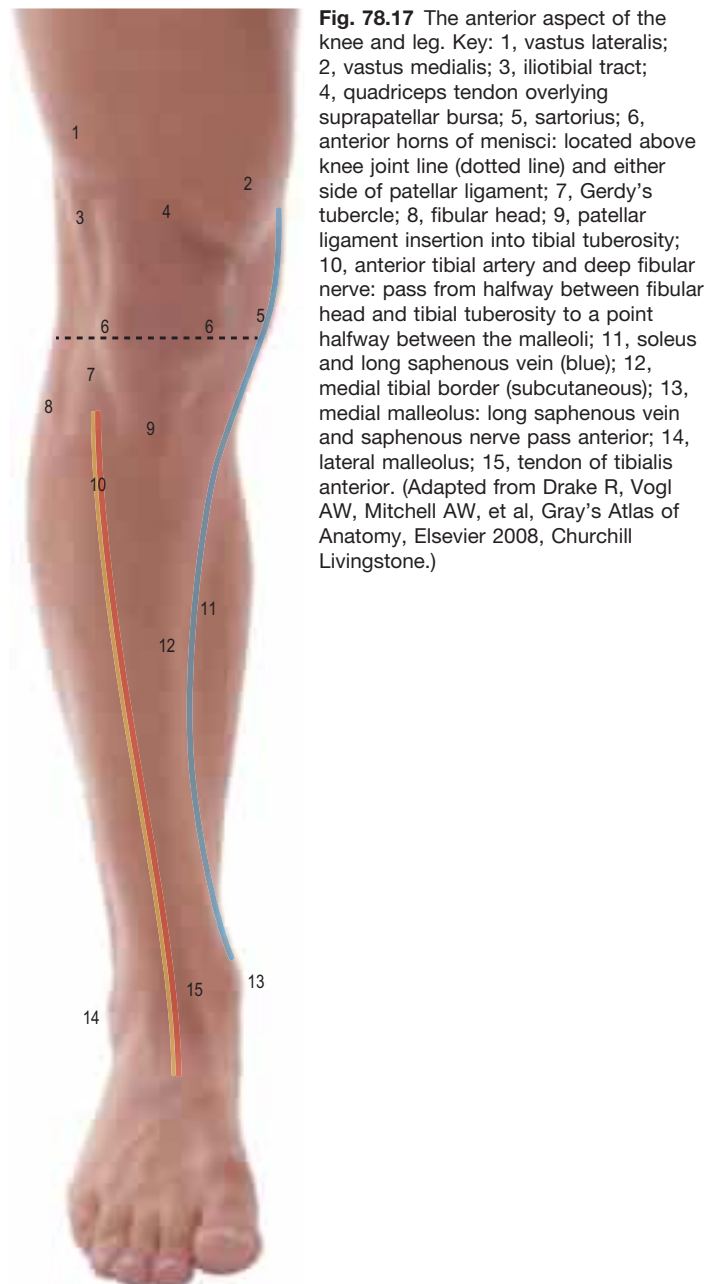


Fig. 78.17 The anterior aspect of the knee and leg. Key: 1, vastus lateralis; 2, vastus medialis; 3, iliotibial tract; 4, quadriceps tendon overlying suprapatellar bursa; 5, sartorius; 6, anterior horns of menisci: located above knee joint line (dotted line) and either side of patellar ligament; 7, Gerdy's tubercle; 8, fibular head; 9, patellar ligament insertion into tibial tuberosity; 10, anterior tibial artery and deep fibular nerve: pass from halfway between fibular head and tibial tuberosity to a point halfway between the malleoli; 11, soleus and long saphenous vein (blue); 12, medial tibial border (subcutaneous); 13, medial malleolus: long saphenous vein and saphenous nerve pass anterior; 14, lateral malleolus; 15, tendon of tibialis anterior. (Adapted from Drake R, Vogl AW, Mitchell AW, et al, Gray's Atlas of Anatomy, Elsevier 2008, Churchill Livingstone.)

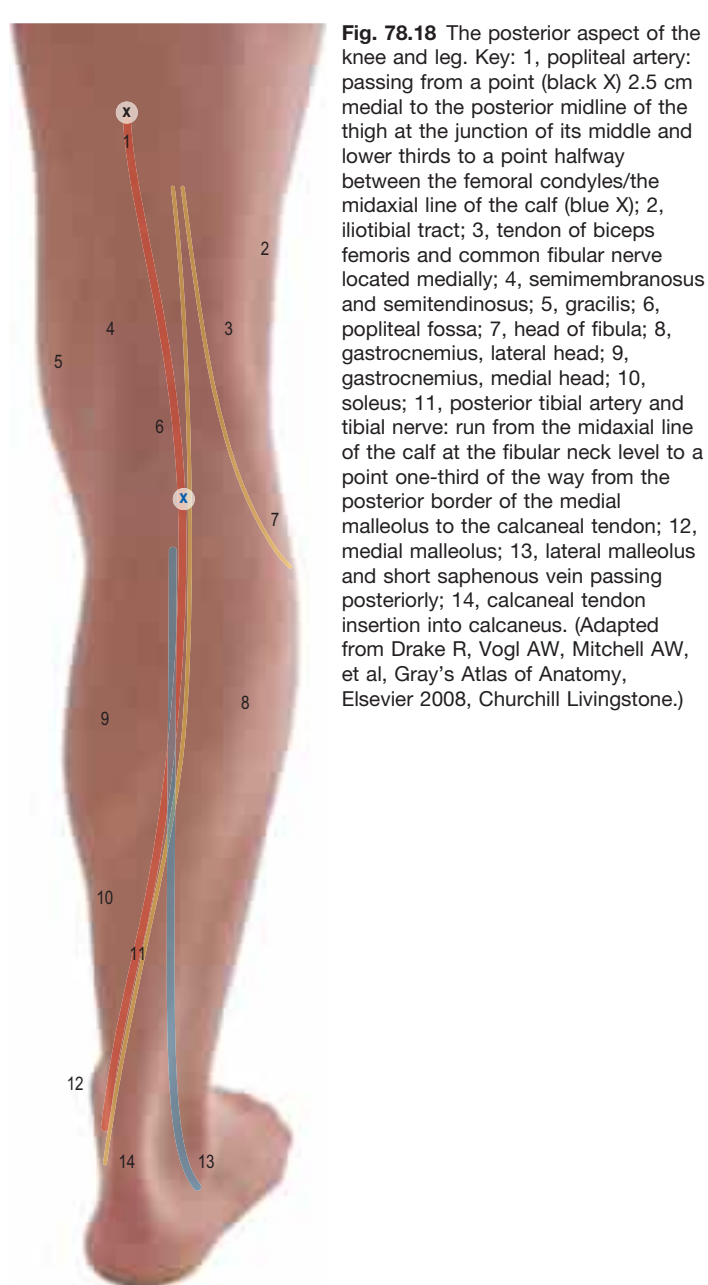


Fig. 78.18 The posterior aspect of the knee and leg. Key: 1, popliteal artery: passing from a point (black X) 2.5 cm medial to the posterior midline of the thigh at the junction of its middle and lower thirds to a point halfway between the femoral condyles/the midaxial line of the calf (blue X); 2, iliotibial tract; 3, tendon of biceps femoris and common fibular nerve located medially; 4, semimembranosus and semitendinosus; 5, gracilis; 6, popliteal fossa; 7, head of fibula; 8, gastrocnemius, lateral head; 9, gastrocnemius, medial head; 10, soleus; 11, posterior tibial artery and tibial nerve: run from the midaxial line of the calf at the fibular neck level to a point one-third of the way from the posterior border of the medial malleolus to the calcaneal tendon; 12, medial malleolus; 13, lateral malleolus and short saphenous vein passing posteriorly; 14, calcaneal tendon insertion into calcaneus. (Adapted from Drake R, Vogl AW, Mitchell AW, et al, Gray's Atlas of Anatomy, Elsevier 2008, Churchill Livingstone.)

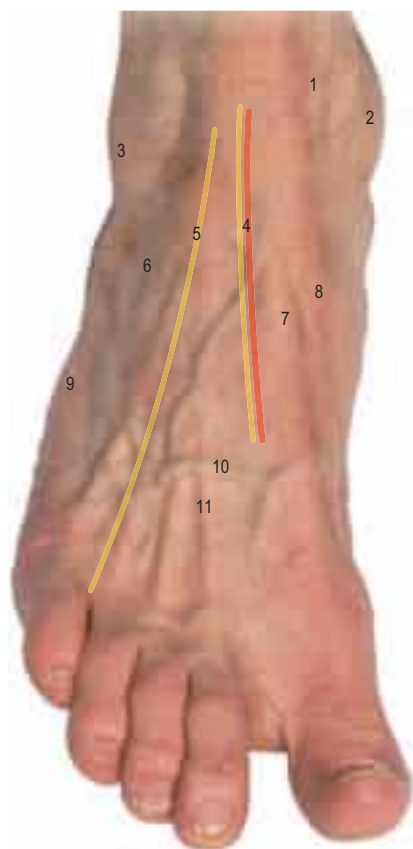


Fig. 78.19 The dorsal aspect of the foot and ankle. Key: 1, long saphenous vein and saphenous nerve; 2, medial malleolus; 3, lateral malleolus; 4, dorsalis pedis artery and deep fibular nerve: located immediately lateral to the tendon of extensor hallucis longus; 5, intermediate dorsal cutaneous nerve: visible passing towards the fourth digit during digit and ankle plantar flexion; 6, extensor digitorum brevis; 7, tendon of extensor hallucis longus; 8, tendon of tibialis anterior: insertion into medial cuneiform; 9, tuberosity of fifth metatarsal; 10, dorsal venous arch of foot; 11, tendon of extensor digitorum longus (to digit two). (Adapted from Drake R, Vogl AW, Mitchell AW, Gray's Anatomy for Students, 2nd ed, Elsevier 2010, Churchill Livingstone.)

Foot

The anterior part of the upper surface of the calcaneus can be identified slightly anterior to the lateral malleolus on the dorsum of the foot. When the foot is passively inverted, the upper and lateral parts of the head of the talus can be both seen and palpated approximately 3 cm anterior to the distal end of the tibia; the extensor tendons obscure them when the toes are extended. Regional tenderness can indicate a talar neck fracture.

The dorsal aspects of the bodies of the metatarsals can be felt more or less distinctly, although they are partly obscured by the extensor tendons of the toes. The tuberosity on the base of the fifth metatarsal forms a distinct palpable and visible projection on the lateral border of the foot.

The flat lateral surface of the calcaneus can be palpated on the lateral aspect of the heel and can be traced anteriorly and inferior to the lateral malleolus, where the tendons of fibularis longus and brevis hide it. The fibular (peroneal) tubercle, when sufficiently large, can be palpated 2 cm inferior to the tip of the lateral malleolus. A palpable depression just anterior to the lateral malleolus leads to the lateral end of the tarsal sinus.

On the medial side of the foot, the sustentaculum tali can be felt approximately 2 cm inferior to the medial malleolus, with the tendon of flexor hallucis longus located inferiorly (Fig. 78.20). The navicular tuberosity is palpable and usually visible approximately 2.5 cm anterior to the sustentaculum tali. Anterior to this, the medial cuneiform can be identified, with the tendon of tibialis anterior inserting on to it. The upper and medial parts of the joint between the medial cuneiform and the first metatarsal can be felt as a narrow groove.

When the foot is placed on the ground, it rests on the posterior part of the inferior surface of the calcaneus, the heads of the metatarsals and, to a lesser extent, on its lateral border. The medial longitudinal arch of the foot, the instep, is elevated from the ground. The medial and lateral

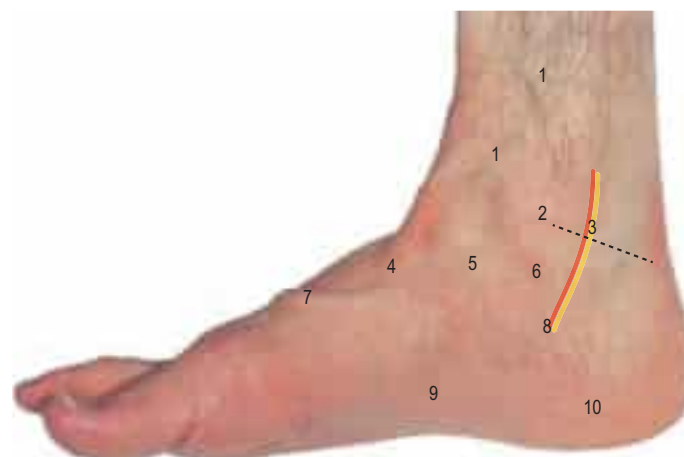


Fig. 78.20 The medial aspect of the foot and ankle. Key: 1, long saphenous vein and saphenous nerve; 2, medial malleolus; 3, posterior tibial artery and tibial nerve: located approximately one-third of the distance along a line passing from the posterior border of the medial malleolus to the calcaneal tendon; 4, tendon of tibialis anterior: insertion into medial cuneiform; 5, navicular tuberosity and tibialis posterior insertion; 6, sustentaculum tali; 7, dorsal venous arch; 8, region of the posterior tibial artery bifurcation into medial and lateral plantar arteries; 9, medial border of plantar aponeurosis; 10, medial calcaneal tubercle covered by fibro-fatty heel pad. (Adapted from Drake R, Vogl AW, Mitchell AW, Gray's Anatomy for Students, 2nd ed, Elsevier 2010, Churchill Livingstone.)



Fig. 78.21 The anterolateral aspect of the foot and ankle. Key: 1, tendons of fibularis longus and brevis; 2, tendons of extensor digitorum longus and fibularis tertius; 3, inferior extensor retinaculum; 4, lateral malleolus; 5, fibular tubercle/trochlea; 6, tendon of fibularis tertius; 7, calcaneocuboid joint and transverse tarsal joint line; 8, tuberosity of fifth metatarsal and tarsometatarsal (Lisfranc) joint line (dotted). (Adapted from Drake R, Vogl AW, Mitchell AW, Gray's Anatomy for Students, 2nd ed, Elsevier 2010, Churchill Livingstone.)

tubercles of the calcaneus can be palpated on its inferior surface via the anterior region of the fibro-fatty heel pad, which covers and obscures them. The heads of the metatarsals are similarly covered by a fibro-fatty pad on the plantar surface of the foot and can be palpated on the dorsal foot, with the digits in flexion.

The calcaneocuboid joint is about 2 cm posterior to the tuberosity on the base of the fifth metatarsal and is almost in line with the talonavicular joint, whose position may be gauged from the talar head (Fig. 78.21). The tarsometatarsal (Lisfranc) joints lie on a line joining the tuberosity of the fifth metatarsal to the tarsometatarsal joint of the great toe, which is located 2–3 cm anterior to the navicular tuberosity and is palpable. The joint between the second metatarsal and the intermediate cuneiform is located 2–3 mm posterior to the line of the other

tarsometatarsal joints. The metatarsophalangeal joints lie about 2.5 cm behind the web spaces of the toes and are palpable on the dorsal foot, where they are accessible via needle.

MUSCULOTENDINOUS AND LIGAMENOUS LANDMARKS



See Video 78.1.

Buttock and hip

The bulky prominence of the buttock is formed by gluteus maximus, a variable amount of subcutaneous fat and the normal alignment of the pelvis, which angles the ischium posteriorly (see Fig. 78.16).

The horizontal gluteal fold marks the upper limit of the posterior aspect of the thigh. It does not correspond to the lower border of gluteus maximus but is formed by fibrous connections between the skin and the deep fascia. The intergluteal cleft, which separates the buttocks inferiorly, begins superiorly at the S3 or S4 vertebrae. The superior border of gluteus maximus begins on the iliac crest 5–7 cm superolateral to the posterior superior iliac spine and runs inferiorly and laterally towards the apex of the greater trochanter. Its lower border corresponds to a line drawn from the ischial tuberosity, through the midpoint of the gluteal fold, to a point approximately 9 cm below the greater trochanter. Although gluteus maximus overlaps the ischial tuberosity in the standing position, on sitting, it slides superiorly posterior to the tuberosity, leaving it free to bear weight. Gluteus maximus can be felt to contract when the hip is extended against resistance.

Gluteus medius completely covers the underlying gluteus minimus. Both muscles lie in a slight depression superolateral to gluteus maximus and inferior to the anterior portion of the iliac crest and both pass inferiorly to insert into the greater trochanter. They constitute the major abductors of the hip and are demonstrated by asking the subject to stand on one limb. The ipsilateral muscles contract to stabilize the centre of gravity and maintain a relatively horizontal pelvic position. Paresis or paralysis causes pelvic tilt towards the unsupported, contralateral side (Trendelenburg's sign).

Thigh

The inguinal ligament can be felt running between the anterior superior iliac spine and the pubic tubercle, especially when the thigh is abducted and laterally rotated (see Fig. 78.15). The groin/inguinal crease (Holden's line), where the deep layer of the subcutaneous tissue of the anterior abdominal wall meets the fascia lata of the thigh, is distal to the inguinal ligament and runs obliquely. It lies a mean of 6.7 ± 1.9 cm distal to the inguinal ligament (Lechner et al 1988).

The femoral triangle is located on the proximal anterior thigh. It is bounded superiorly by the inguinal ligament and laterally by the strap-like sartorius, which can be both seen and felt in a reasonably thin and muscular subject when the hip is flexed in the sitting position, while the knee is kept extended and the thigh is slightly abducted and rotated laterally. Sartorius can be traced inferomedially from the anterior superior iliac spine to approximately halfway down the medial side of the thigh; distally, it may be identified as a soft longitudinal ridge passing towards the posterior part of the medial femoral condyle. The adductor group of muscles forms the bulky, fleshy mass at the upper part of the medial thigh. The medial boundary of adductor longus forms the medial boundary of the femoral triangle and can be felt as a distinct ridge when the thigh is adducted against resistance. At its superior end, its prominent tendon of origin can be seen and palpated immediately inferior to the pubic tubercle, which is a useful guide to this bony landmark.

The forward convexity of the anterior thigh is caused by the curvature of the femur covered by the muscle mass of quadriceps femoris. Rectus femoris appears as a raised ridge passing down the anterior aspect of the thigh to the patellar base, when the sitting subject flexes the hip with the knee extended. Vastus medialis constitutes a bulge that is both superior and medial to the patella (see Fig. 78.17). Vastus lateralis forms an elevation superior and lateral to the patella that is more proximal and less pronounced than that of vastus medialis. Vastus intermedius is covered by the three other muscles in the group. The thick tendon of adductor magnus can be palpated on the distal medial thigh, deep within the indentation formed between vastus medialis anteriorly and gracilis and sartorius posteriorly.

The iliotibial tract, a thickened portion of the deep fascia of the thigh (fascia lata), runs from the lateral aspect of the iliac crest to Gerdy's

tubercle and produces the flattened appearance of the lateral thigh. It forms a defined ridge that is visible on the lateral aspect of the knee either when the knee is extended against gravity or when the opposite limb is lifted from the floor while standing.

The muscles of the posterior thigh may be palpated from the ischial tuberosity inferiorly along the lateral (biceps femoris) and medial (semitendinosus and semimembranosus) aspects of the thigh.

Knee

The large depression visible posterior to the knee when the joint is actively flexed against resistance corresponds to the popliteal fossa. The transverse skin crease of the popliteal fossa is located 2–3 cm superior to the knee joint line. The fossa is bounded on the superolateral side by the prominent tendon of biceps femoris, which can be palpated between the finger and thumb and traced inferiorly to the head of the fibula. Three tendons, of semitendinosus, gracilis and semimembranosus, can be felt on the superomedial side of the fossa. Semitendinosus is the most lateral and posterior, while gracilis is the most medial and anterior; both tendons stand out sharply and can be seen when the knee is flexed against resistance and the limb actively adducted. The tendon of semimembranosus may be palpated deeply in the interval between the tendons of semitendinosus and gracilis. It is much thicker than the other two tendons and broadens rapidly as it is traced proximally. The upper borders of the two heads of gastrocnemius form the inferomedial and inferolateral boundaries of the popliteal fossa.

The cord-like fibular collateral ligament may be felt passing from the apex of the fibular head to the lateral epicondyle of the femur when the knee is flexed and laterally directed pressure (valgus force) is applied to the medial side of the knee. The medial patellar retinaculum may be felt as a flat, broad band overlying the medial femoral condyle in the flexed knee, running between the midpoint of the medial side of the patella and the medial femoral epicondyle.

Leg

The muscles in the anterior osteofascial compartment of the leg form a gentle prominence over the upper two-thirds of its anterolateral aspect; this prominence is accentuated when the foot is actively dorsiflexed. These muscles are replaced by their tendons in the lower third of the leg. The tendon of tibialis anterior can be seen just lateral to the anterior border of the tibia and traced inferiorly and medially across the anterior aspect of the ankle to the medial cuneiform (see Figs 78.17, 78.19). The other tendons cannot be examined satisfactorily superior to the ankle. Immediately superior to the posterior border of the medial malleolus and close to the medial border of the tibia, the tendons of tibialis posterior and flexor digitorum longus can be felt (rather indistinctly) when the foot is actively inverted and plantar flexed.

On the lateral aspect of the leg, fibularis longus can be seen as a narrow ridge following the line of the lateral aspect of the fibula during active eversion and plantar flexion of the foot. It covers and hides fibularis brevis. The shaft of the fibula can only be palpated indistinctly between its neck and the area above the lateral malleolus.

The bulky prominence of the posterior leg is formed by gastrocnemius and soleus, both of which can be identified either when the foot is plantar flexed against resistance, or when the heel is raised from the ground by standing on tiptoes (see Fig. 78.18). The two heads of gastrocnemius unite to form the inferior borders of the popliteal fossa. The medial head of gastrocnemius descends to a more inferior level than the lateral head. Soleus lies deep to gastrocnemius; when tensed, it bulges from under the medial and lateral margins of gastrocnemius, particularly on the lateral side, and its fleshy belly extends to a more distal level. Both muscles end inferiorly in the conspicuous calcaneal tendon, which can be palpated between the finger and thumb and followed inferiorly to its insertion into the posterior aspect of the calcaneus.

Foot

When the toes are actively extended, the bellies of extensor digitorum brevis and extensor hallucis brevis form a small elevation on the dorsal foot, just anterior to the lateral malleolus (see Fig. 78.19). During ankle inversion and extension of the toes, the prominent tendon of tibialis anterior is visible on the medial side of the dorsal foot, passing inferiorly and medially to the medial cuneiform. The tendon of extensor hallucis longus can be identified laterally; the tendons of extensor digitorum longus and fibularis tertius are further lateral as they pass deep

to the inferior extensor retinaculum and immediately anterior to the lateral part of the distal tibia. More distally, the tendons of extensor digitorum longus and fibularis tertius diverge and may be traced to their insertions.

The tendon of tibialis posterior winds posterior to the medial malleolus and then curves anteriorly in the interval between this bony landmark and the sustentaculum tali to reach the tuberosity of the navicular (see Fig. 78.20). The tendon is visible and palpable posterior to the malleolus when the foot is forcibly plantar flexed and inverted. The tendon of flexor digitorum longus lies immediately posterolateral to that of tibialis posterior. It curves forwards inferior to tibialis posterior and lies on the medial aspect of the sustentaculum tali. From there, it passes anteriorly and laterally to the centre of the plantar foot, where it divides into four tendons that pass to the lateral four toes. The tendon of flexor hallucis longus lies inferior to, and grooves, the sustentaculum tali. As it passes towards the great toe, it crosses the line of the flexor digitorum longus opposite the interval between the sustentaculum tali and the tuberosity of the navicular. Abductor hallucis may be seen in some subjects as a fleshy mass along the medial border of the foot, passing from the medial calcaneal tubercle to the first metatarsophalangeal joint.

When the toes are maximally dorsiflexed, the plantar aponeurosis is easily palpated along the medial border of the foot, from just distal to the heel pad as far as the first metatarsophalangeal joint. It originates from the calcaneal tubercles, which may be palpable via the anterior aspect of the heel pad.

VESSELS, PULSES AND NERVES

Arteries

The femoral artery normally enters the anterior thigh at the mid-inguinal point ± 1 cm either side (medial to lateral) (Hale et al 2010) (see Fig. 78.15). The midinguinal point is halfway between the anterior superior iliac spine and the pubic symphysis. The artery normally passes directly anterior to the hip joint; its course may be represented by the upper two-thirds of a line joining its entry point into the thigh with the adductor tubercle, when the thigh is flexed, slightly abducted and rotated laterally. It travels deep to sartorius from the apex of the femoral triangle to the adductor hiatus. The profunda femoris artery (deep artery of the thigh) normally arises from the lateral side of the femoral artery, 6.5 ± 1.5 cm distal to the mid-inguinal point (Choy et al 2013), which is most commonly at or superior to the level of the inguinal/groin crease (Lechner et al 1988).

The popliteal artery may be represented by a line extending from the junction of the middle and lower thirds of the thigh, 2.5 cm medial to its posterior midline, to a point halfway between the femoral condyles, and continuing inferolaterally to the level of the tibial tuberosity, medial to the fibular neck, where it bifurcates into the anterior and posterior tibial arteries (see Fig. 78.18).

The anterior tibial artery may be represented by a line extending from a point halfway between the tibial tuberosity and the fibular head to a point midway between the anterior borders of the two malleoli (see Fig. 78.17). It passes on to the dorsal foot lateral to the tendon of extensor hallucis longus, as the dorsalis pedis artery. The latter gives off a medial branch that travels toward the first intermetatarsal space.

The posterior tibial artery may be represented by a line drawn from a midaxial point on the calf (approximately midway between the tibial condyles) at the level of the neck of the fibula, to a point one-third of the way between the posterior border of the medial malleolus and the calcaneal tendon (see Fig. 78.18). The same line represents the course of the tibial nerve. At first, the nerve lies lateral to the popliteal artery but gradually crosses the vessel to gain its medial side. The posterior tibial artery bifurcates into the plantar arteries halfway between the medial malleolus and the medial calcaneal tubercle (see Fig. 78.20).

The medial plantar artery passes anteriorly towards the great toe. The lateral plantar artery passes obliquely towards a point approximately 2.5 cm medial to the tuberosity of the fifth metatarsal, after which it enters the proximal part of the fourth intermetatarsal space. Digital arteries and nerves pass along the medial and lateral sides of the toes.

Pulses

Femoral artery The pulsations of the femoral artery can be felt at its entry point into the proximal femoral triangle, where it can be compressed against either the superior pubic ramus or the hip joint. The pulse of the femoral artery is of value in assessing whether there is any

significant cardiac output in cases of circulatory collapse. It is a common site for catheter insertion for radiological examinations such as cardiac angiography and for arterial puncture for blood gas analysis.

Popliteal artery The pulse of the popliteal artery is the most difficult of the peripheral pulses to feel because the artery lies deep in the popliteal fossa. It is best examined with the subject lying supine or prone, with the knee flexed in order to relax the tense popliteal fascia that roofs the popliteal fossa. The popliteal pulse is then felt over the midline of the fossa by deep pressure against the popliteal surface of the femur.

Posterior tibial artery The pulse of the posterior tibial artery can be felt by gentle palpation posterior to the medial malleolus as the artery lies between the tendons of flexor hallucis longus and flexor digitorum longus, approximately one-third of the way along a line connecting the posterior border of the medial malleolus to the calcaneal tendon.

Dorsalis pedis artery The dorsalis pedis arterial pulse is found by palpation against the underlying tarsals, immediately lateral to the tendon of extensor hallucis longus, from the midpoint between the malleoli to the proximal end of the first intermetatarsal space.

Veins

The femoral vein is located immediately medial to the femoral artery pulse, where it can be accessed (see Fig. 78.15). While femoral vein puncture is relatively easy and supplies ready access to the right atrium, the use of this approach is relatively unpopular for long-term cannulation because of a greater incidence of thrombosis and sepsis. However, it is a useful site for venous sampling in a patient with collapsed veins. For femoral vein cannulation, the skin puncture site is approximately 1 cm medial to the femoral artery and just inferior to the inguinal ligament.

The dorsal venous arch of the foot is visible and curves across the metatarsals (see Fig. 78.19). The long saphenous vein arises from the medial end of the arch, and the short saphenous vein arises from the lateral end. The long saphenous vein can be seen and palpated anterior to the medial malleolus, after which it runs superiorly and posteriorly across the distal medial surface of the tibia: this is a useful site for surgical venous access (cut-down) or venous harvesting for cardiac bypass surgery (see Figs 78.17, 78.19). The vein then runs proximally along the posterior aspect of the medial border of the tibia, where it is accompanied by the saphenous nerve, to reach a point a hand's breadth (7–10 cm) posterior to the medial patellar border. From here, the vein ascends the thigh to the saphenous opening, where it passes deep to enter the femoral vein. An image-based study found that the saphenofemoral junction almost always occupied a 3×3 cm zone located 1–4 cm lateral and 0–3 cm inferior to the pubic tubercle (Mirjalili et al 2014) (see Fig. 78.15). The junction was nearer to the pubic tubercle in younger and thinner subjects.

The course of the short saphenous vein may be represented by a line from the posterior surface of the lateral malleolus superiorly along the midline of the calf to the popliteal fossa; together with the sural nerve, this vein is the key anatomical guide to surgical dissection of the popliteal fossa (see Fig. 78.18).

Nerves

In the proximal femoral triangle, the femoral nerve is located immediately lateral to the femoral artery pulse.

The surface marking for the course of the saphenous nerve in the leg matches that of the long saphenous vein (see above). The lateral femoral cutaneous nerve enters the femoral triangle 2.1–3.5 cm (range 0.3–7.3 cm) from the anterior superior iliac spine (see Fig. 78.15) (Grothaus et al 2005, Bjurlin et al 2007, Ropars et al 2009). It then passes over the lateral border of sartorius up to 11.3 cm distal to the anterior superior iliac spine (Grothaus et al 2005, Ropars et al 2009), and over the anterior border of tensor fasciae latae between 2.4 and 9.2 cm distal to the anterior superior iliac spine (Ropars et al 2009). The nerve is at risk of damage during anterolateral surgical approaches to the hip.

The course of the sciatic nerve can be represented by a curved line that starts at a point 0.3 ± 0.05 of the distance down a line drawn from the posterior superior iliac spine to the ischial tuberosity, and passes to a point 0.5 ± 0.05 of the distance along a line drawn from the ischial tuberosity to the greater trochanter (see Fig. 78.16). It then continues vertically and inferiorly in the midline of the posterior aspect of the

thigh, deep to biceps femoris, to the upper angle of the popliteal fossa, where it divides into the tibial and common fibular nerves (if it has not already done so at a more superior level).

The surface marking for the course of the tibial nerve in the leg matches that of the posterior tibial artery (see above). The course of the sural nerve at the ankle can be represented by a line running parallel to the calcaneal tendon that is situated halfway between the tendon and the lateral malleolus: this position is variable and the sural nerve is at risk from surgery in this region.

The course of the common fibular nerve can be indicated by a line that runs from the superior angle of the popliteal fossa, along the medial side of the tendon of biceps femoris (see Fig. 78.18) and then curves inferiorly and anteriorly around the neck of the fibula approximately 3 cm distal to the superior tip of the fibula. The common fibular nerve is palpable medial and distal to the tendon of biceps femoris and over the neck of the fibula, although here it becomes less distinct as it passes deep to the origin of fibularis longus. At the neck of the fibula, the nerve is at particular risk of damage either from a tightly applied plaster cast or from a fracture.

The deep fibular nerve starts on the lateral aspect of the neck of the fibula, passes downwards and medially, and rapidly becomes associated with the anterior tibial artery, with which it is landmarked (see Fig. 78.18). The superficial fibular nerve also begins on the lateral aspect of the neck of the fibula. It descends to a point on the anterior border of fibularis longus, at the junction of the middle and lower thirds of the leg, where it pierces the deep fascia and divides into intermediate and medial dorsal cutaneous branches. The intermediate dorsal cutaneous branch is normally both visible and palpable over the dorsolateral aspect of the ankle and foot toward the fourth toe (see Fig. 78.19). The medial dorsal cutaneous branch is rarely visible and passes towards the second toe. These superficial nerves are at risk from surgery in the anterior ankle region (especially during arthroscopy).

CLINICAL PROCEDURES

Nerve anaesthesia (nerve blocks)

Nerves can be effectively anaesthetized with local anaesthetic injection for surgical or post-injury pain relief. Common injection sites for the lower limb are around the hip for thigh and knee pain, and around the ankle for foot surgery. For the deeper nerves around the hip, the use of a nerve stimulator is very helpful in localizing the target nerve precisely before infiltration of local anaesthetic. For example, blockade of the sciatic nerve in the buttock, femoral nerve in the anterior thigh, and obturator nerve in the medial thigh may be undertaken to provide pain relief for total knee replacement; the tibial nerve at the posteromedial ankle, the sural nerve at the posterolateral ankle and the superficial fibular nerves at the anterior ankle may all be anaesthetized at the ankle to allow 'awake foot surgery'.

Intramuscular injection

Intramuscular injections into the buttock should be avoided to prevent iatrogenic damage to the sciatic nerve. If the buttock is to be used, a relatively safe area is the true upper and outer quadrant, which is identified with the whole buttock exposed. The injection is then given mainly into gluteus medius rather than into gluteus maximus, provided a sufficiently long needle is used; most so-called 'intramuscular' injections given into the buttock are actually given into the fat. A safe alternative is to inject into the lateral aspect of the thigh (vastus lateralis).

Joint injection and aspiration

Careful aseptic technique is essential for all joint aspirations and injections.

Hip Intra-articular hip injections are typically performed under fluoroscopic or ultrasonographic guidance. With the patient lying supine, and after the positions of the femoral artery and the anterior superior iliac spine have been marked out, the needle is introduced anteriorly, approximately 5 cm distal to the anterior superior iliac spine and 4 cm lateral to the femoral artery pulse, and passed posteriorly, a little proximally and medially. An alternative technique that is dependent on relative distances from surface landmarks, rather than imaging guidance, utilizes the site of the proximal anterolateral portal for hip arthroscopy as the point of needle insertion (Masoud and Said 2013).

Knee The lateral retropatellar approach is used. With the patient supine and the knee extended, the needle is introduced at the level of the superior border of the patella and guided towards the suprapatellar bursa.

Popliteal fossa Sciatic nerve blocks at the popliteal fossa in infants and children can be effectively performed above the popliteal skin crease by correcting the adult landmark of 10 cm by the ratio of the calculated child femoral shaft length over the adult femoral shaft length (Bernière et al 2008).

Ankle The anterior approach entails introducing the needle between the tendons of tibialis anterior and extensor hallucis longus with the foot partially plantar flexed.

Arthroscopy portals

The placement of portals for arthroscopy is important for maximizing surgical access for visualization and for surgical instruments, and also for avoiding damage to structures such as nerves and blood vessels.

Hip Arthroscopy of the hip is not yet a common procedure but is increasingly being used. A variety of entry portals have been described. The anterolateral portal is sited on the skin approximately 4 cm lateral to the femoral artery pulse and 4 cm inferior to the inguinal ligament. Under fluoroscopy, a needle traversing the skin 30–45 degrees proximally is advanced into the joint. Other more lateral portals can be used; the lateral femoral cutaneous nerve and the sciatic nerve are potentially at risk.

Knee The knee is the joint most frequently examined with arthroscopy. The standard portals are placed anteriorly. When the knee is flexed at a right angle, 'soft triangles' bordered by the patellar ligament, the femoral condyles and the tibia and anterior horns of the menisci are palpable on either side of the superior third of the patellar ligament. Small incisions can be made in the apices of these triangular areas at about the level of the inferior pole of the patella for the anterolateral portal, and slightly lower for the anteromedial portal. These portals will allow passage of an arthroscope and instruments, with good access to most of the joint. The patellar ligament and the infrapatellar branch of the saphenous nerve are at risk from the skin incisions. The nerve is less vulnerable than the ligament; its position is variable but it usually lies below the appropriate portal sites. In the days of open meniscal surgery, the nerve was usually divided and painful neuroma formation was not uncommon.

Posteromedial and posterolateral portals are useful when better access to the posterior knee is required. The saphenous and common fibular nerves, respectively, are at risk with these approaches, and anatomical knowledge is vital to ensure safe portal placement. Laterally, the incision should be anterior to the tendon of biceps femoris, which can easily be palpated. Medially, the situation is more difficult because the saphenous nerve and long saphenous vein travel superficial to the medial epicondylar region of the knee, approximately a hand's breadth posterior to the medial border of the patella. It is essential that sharp incision includes the skin only, and that dissection down to the joint capsule is undertaken bluntly. If the medial meniscus is sutured, the capsule must be exposed before tying off sutures to avoid ensnaring the nerve.

Ankle The use of ankle arthroscopy is well established, especially in treating sports-related injuries. Anterior portals are standard. The anteromedial portal can be placed just medial to the tendon of tibialis anterior, in the palpable soft spot. However, this comes close to the long saphenous vein and saphenous nerve; limiting the sharp incision to the skin, followed by blunt deeper dissection, reduces the risk. Alternatively, the anteromedial portal can be chosen to pass between the lateral edge of tibialis anterior and extensor hallucis longus. The anterolateral portal is placed with the help of the arthroscope in the joint to check the position of a preliminary needle passed into it. The portal should pass lateral to fibularis tertius and extensor digitorum longus. The intermediate dorsal cutaneous branch of the superficial fibular nerve is at risk as it crosses the ankle anterior to the lateral malleolus, and therefore it is wise to mark the positions of the nerves and blood vessels with a pen before making any incision. The nerves can usually be felt at the front of the ankle, even in obese patients. They are often seen if the ankle is maximally pulled into plantar flexion. Posterior ankle arthroscopy is controversial: there is a view that the proximity of the neurovascular bundle (medially) and the sural nerve (laterally) renders it too

hazardous. Keeping lateral to the tendon of flexor hallucis longus will safeguard the medial neurovascular bundle.

Placement of surgical incisions

Hip Surgery of the hip joint is usually undertaken for congenital issues, slipped capital femoral epiphysis in children, trauma or arthroplasty. Anterior and anterolateral approaches, often used in children, put the lateral femoral cutaneous nerve at considerable risk. Even in more lateral approaches, the lateral femoral cutaneous nerve and the femoral and sciatic nerves are all at risk through traction. A popular anterolateral approach to the hip joint involves splitting and separating forwards the anterior part of gluteus medius and vastus lateralis as a single sheet of tissue for subsequent reattachment to the greater trochanter. This technique relies on the anatomical continuity of the tissue. If the splitting of gluteus medius is more than a few centimetres superior to the tip of the greater trochanter, then the superior gluteal nerve and vessels are at risk, and weakness of hip abduction and a limp may result. In the posterior approach to the hip joint, gluteus maximus is incised in the line of its fibres. This is followed by division of the tendons of the short lateral rotators (piriformis, obturator internus and externus and gemelli) near their trochanteric attachments to reveal the joint capsule. Reflecting these cut tendons medially protects the sciatic nerve. The inferior and superior gluteal nerves are at risk during posterior approaches to the hip.

Knee Most open knee surgery can be undertaken through an anterior midline longitudinal incision, which gives good access and means that any future surgery can usually be undertaken via the same wound. New incisions run the risk of skin necrosis and poor wound healing as a consequence of interfering with the cutaneous blood supply. Inevitable interruption of cutaneous nerves, including the infrapatellar branch of

the saphenous nerve, means that there is always numbness lateral to a longitudinal incision.

The extensile approach to the posterior knee is extensive. The key is to expose the sural nerve and short saphenous vein and to trace them proximally. This will lead the surgeon into the popliteal fossa and, after opening the deep fascia, safely to the neurovascular bundle. The wound crosses a flexure crease. A scar perpendicular to the crease might induce a fixed flexion contracture of the joint and therefore an S-shaped incision is employed where the transverse segment runs in the line of the flexure crease.

Ankle and foot Incisions around the foot and ankle frequently put cutaneous nerves at risk and such an injury can result in a neuroma. The sural nerve at the ankle has a notorious tendency to form neuromas if transected, often after repair of a ruptured calcaneal tendon.



Bonus e-book tables and video

Table 78.1 Segmental innervation of the muscles of the lower limb.

Table 78.2 Segmental innervation of joint movements of the lower limb.

Table 78.3 Movements, muscles and segmental innervation in the lower limb.

Table 78.4 The movements and muscles tested to determine the location of a lesion in the lower limb.

Video 78.1 Lower limb surface anatomy.

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